When Does Trade Protection Help? Evidence from the Rise of American Manufacturing

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Abstract

When does trade protection boost growth and when does it harm growth? I study this question in the context of US industrialization in the 19th century. Using newly digitized tariff, import price, and railway freight data, I develop a novel empirical design leveraging county-product variation in exposure to nationwide tariffs based on distance to international ports. Exploiting a nationwide fiscal shock that suddenly raised tariffs for reasons unrelated to local industry conditions, I find protection reduces output growth for products close to the technological frontier but increases growth for products far behind it. I also find protection shifts patent activity towards far-from-frontier products, suggesting a reshaping of comparative advantage. I present a model of trade and innovation with variable markups generating these empirical results: for close-to-frontier products, protection substitutes for innovation as a source of market power; for further-behind products, protection instead makes the market contestable enough that the domestic producer can capture the rents from innovation. This model provides an imperfect-competition rationale for infant industry protection distinct from the usual learning-by-doing justification. I conclude by studying the welfare effects of trade, which can be positive or negative in contrast to workhorse trade models.

*University of California, San Diego. I am indebted to Sam Bazzi, Fabian Eckert, Fabian Trottner, Munseob Lee, and Marc Muendler for their constant support and guidance. I thank Reka Juhasz, Juan Herreno, Martin Rotemberg, Harrison Mitchell, Benny Kleinman, Shinnosuke Kikuchi, Jan Bakker, Ezra Oberfield and Roman Zarate for valuable comments that improved the paper.

1 Introduction

One of the oldest ideas in all of economics is the infant industry argument: trade protection insulates new domestic producers from foreign competition, thereby allowing these producers to catch up to the international technological frontier (Hamilton 1791, List 1841, Mill 1848). This claim has proven to be remarkably durable, yet empirical evidence on its effects remains mixed. In the 19th century US context, Head (1994), Baack and Ray (1974), Sundararajan (1970) and Harley (1992) argue trade protection successfully led to the development of the steel rail and cotton textile industries. Others such as Irwin (2000a, 2000b) and Irwin and Temin (2001) argue tariffs had no major impact on the development of the tinplate, pig iron and cotton textile sectors. Klein and Meissner (2024) and Taussig (1915, 1931) argue positive effects for some sectors but null or negative effects for others. The broader industrial policy literature (Lashkaripour and Wu 2025) and trade-and-innovation literature (Shu and Steinwender 2019) also find mixed effects depending on policy, context and sector.

I revisit the 19th century US setting with newly digitized historical data and a novel empirical design, addressing pervasive endogeneity concerns surrounding policymaker targeting of sectors for protection (Juhasz, Lane and Rodrik 2024). In contrast to past studies, which tend to focus on a single product or sector, I look at multiple products jointly. I find that a product's distance to the technological frontier determines the sign of the effect of protection, rationalizing some of the disagreement in the literature. More specifically, I find that protection reduces growth for products sufficiently close to the frontier but raises it for those sufficiently far behind. I develop a model of trade and innovation with variable markups that generates this pattern, and study the welfare and policy implications of trade under this model.

My empirical design is based on the insight that, much like a tariff discourages consumption of imports by raising their prices, high transportation costs to international ports also serve as protection for interior counties. When nationwide tariffs go up, areas near international ports see larger proportional increases in protection than interior areas, which are already shielded from imports by high transportation costs. Methodologically, this approach allows for an across-county, within-product comparison, which is infeasible in canonical approaches. For example, when nationwide steel tariffs rise, Philadelphia steel producers — located at a major international port — see a larger proportional increase in protection than Chicago steel producers, located in the interior. Comparing their growth rates isolates the effect of protection from any nationwide trends in the steel industry, such as technological developments.[1] By contrast, a large empirical trade literature exploits cross-product variation by comparing products that received higher tariff cuts to those that received smaller cuts after a nationwide trade liberalization episode (e.g. Pavcnik 2002, Khandelwal and Topalova 2011, De Loecker et al. 2016, Fajgelbaum et al. 2020). However, product level-trends correlated with tariffs (whether coincidental or targeted by policymakers) would be a concern for this design – one that my within-product comparison addresses via product fixed effects. Shift-share approaches (e.g. Autor, Dorn and Hanson 2013) are also subject to this concern, as Borusyak, Hull and Jaravel (2022) show.[2] The local labor market approach leverages differences in local industrial composition: counties are more exposed when their baseline employment is concentrated in products hit by larger national shocks. My design instead generates spatial variation in protection across counties for the same product. Unlike shift-share designs, my approach allows me to absorb national product-level shocks, such as technological breakthroughs, with product fixed effects; Appendix B.6 formally compares the two designs in more detail.

A second benefit of the within-product, across-space design is that I can identify product-specific treatment effects, in contrast to both the shift-share and cross-product approach, which at best can only identify a convex-weighted average of treatment effects across prod-

[1] For example, the Bessemer process, invented in 1856, enabled the mass production of steel for the first time in human history.

[2] This is true for designs based on the "exogenous shocks" view of Borusyak, Hull and Jaravel (2022). The "exogenous shares" view of Goldsmith-Pinkham, Sorkin and Swift (2020) requires that industrial composition is uncorrelated with future trends. This is unlikely to hold in my context, as locations specializing in different sectors are likely to be exposed to different shocks such as immigrant flows, technological changes and unionization drives. See discussion in Borusyak, Hull and Jaravel (2025).

ucts. The product-specific treatment effect is arguably the more useful causal parameter as policymakers can set tariffs according to product or sector. In addition, the industrial policy literature stresses that the effects of trade protection vary across products or sectors (e.g. Lashkaripour and Wu 2025, Bartelme et al. 2025).

The 1860s provide a useful natural experiment: With tariffs accounting for 90% of Federal revenues (Irwin 2020), Civil War fiscal pressures drove rapid tariff increases, with Congress raising tariffs five times between 1861 and 1865. In addition, US trading partners did not retaliate, allowing me to isolate the import protection channel. Identification requires that nationwide tariff changes are uncorrelated with idiosyncratic county-product shocks. For this condition to be violated, policymakers would need to identify and target idiosyncratic county-product trends across roughly 1,500 counties and hundreds of tariff line items—an informational and cognitive burden far beyond what 19th-century wartime legislators could satisfy, let alone build Congressional consensus around. I validate this narrative by showing that county-product pairs receiving more protection are balanced on baseline observable characteristics relative to those receiving less. Finally, I employ a battery of robustness checks to control for wartime destruction, Emancipation, Reconstruction, military procurement, lobbying power and input disruptions.

Operationalizing this design requires combining several historical data sources, several of which I newly digitize. I digitize product-level import prices and tariffs from the Foreign Commerce and Navigation Tables of the United States for 1859 and 1869. To empirically discipline the functional form of transportation costs in this period, I digitize commodity-specific internal freight rates from the Aldrich Report (1893). I then use this information with Donaldson and Hornbeck's (2016) transportation network to construct county-level transportation costs to port. To control for changes in the cost of imported inputs, I construct an input-output table using Atack, Bateman and Weiss' (2006) state-level census of manufactures. Finally, I harmonize British and American patent records in the 1850s to construct a product-level measure of distance to the technological frontier.

I find sharply heterogeneous effects of protection depending on a product's distance to the British technological frontier — Britain being the industrial leader of this period.[3] Using UK-US patent ratios in the pre-treatment period as a measure of distance to frontier, protection reduces output growth for products close to the frontier but boosts growth for products far behind it. Turning to innovation, I find that protection shifts patent activity towards laggard products, suggesting a reshaping of comparative advantage.

Finally, I develop a model of trade and innovation with variable markups rationalizing these empirical results. Protection has two effects: a positive substitution effect that shifts demand toward domestic varieties, and an ambiguous innovation effect operating through its impact on the returns to innovation. For sufficiently close-to-frontier products, the domestic producer already commands a large market share and faces sufficiently inelastic residual demand that protection reduces the incremental gain from innovation. In some scenarios, it is profit-maximizing to cut costs associated with innovation and cede market share. If this innovation force is large enough, it can overtake the positive substitution effect and reduce output on net. For some products further behind the frontier, the innovation effect instead amplifies the positive substitution effect. Protection makes the market contestable enough that the domestic producer can capture the returns from innovating; thus, unlike close-to-frontier products, innovation and protection are complements, not substitutes.

This model has a number of interesting implications, such as markups potentially *falling* after a trade protection episode, and the possibility of markups covarying positively *or* negatively with innovation. In addition, it provides an imperfect-competition rationale for infant-industry protection that is distinct from the usual external economy of scale or learning-by-doing arguments[4] emphasized in the classic debate: for products sufficiently behind the

[3] A position it occupied since at least the First Industrial Revolution in the 1760s. See Section 2 for further context.

[4] In his famous critique, Baldwin (1969) argues that learning-by-doing justifies protection only if some additional distortion prevents firms from financing or appropriating future learning gains. If future productivity gains are privately appropriable, firms should be able to borrow against them in complete capital markets; if they are not, the appropriability problem would not necessarily be resolved with protection. By contrast, the mechanism in my model requires only imperfect competition in output markets.

frontier, protection can make the market sufficiently contestable for domestic firms to capture the returns to innovation. Additionally, this model can generate a *decrease* in output from a unilateral increase in protection, even in partial equilibrium. This is in contrast to standard gravity frameworks, which predict that unilateral protection increases (i.e. absent retaliatory responses by partners) always weakly increases output in partial equilibrium. In my model, this occurs when protection reduces innovation effort sufficiently to overwhelm the mechanical substitution effect.

Finally, under a general equilibrium version of the model, the welfare gains from trade are ambiguous, in contrast to standard gravity frameworks which predict positive welfare gains from trade. The model's welfare formula nests the classic ACR formula (Arkolakis, Costinot and Rodríguez-Clare 2012), with an additional ex-post statistic — the change in innovation — governing the sign and magnitude of the welfare effect.

Literature Review: My paper is connected to four different literatures. First, this paper contributes to the literature analyzing the role of trade protection in the development of American manufacturing. Head (1994), Baack and Ray (1974), Sundararajan (1970) and Harley (1992) argue trade protection successfully led to the development of the steel rail and cotton textile industries. Others such as Irwin (2000a, 2000b) and Irwin and Temin (2001) argue tariffs had no major impact on the development of the tinplate, pig iron and cotton textile sectors. Greenland and Lopresti (2024) find trade protection led to the reallocation of labor towards manufacturing. Klein and Meissner (2024) find positive impacts on output for textiles and metals, null effects for chemicals and glass, and negative effects for paper and transportation equipment. Taussig (1915, 1931) suggested that tariffs boosted the development of silk textiles, steel rails and plate glass while having an insignificant impact on cotton textiles and iron and steel. I revisit this topic with new granular data and a novel empirical design, as well as a systematic framework that can account for why some products see a positive effect and why others see a null or negative effect.

Second, this paper contributes to the empirical trade literature by offering a new within-product empirical design to study the impact of trade protection within a country. This contrasts with a large empirical literature relying on cross-sectoral variation in tariff changes – comparing sectors receiving large versus small tariff cuts after a nationwide trade liberalization episode – such as Pavcnik (2002), Khandelwal and Topalova (2011), De Loecker et al. (2016), and Fajgelbaum et al. (2020). Shift-share approaches based on the ”exogenous shocks” view also implicitly rely on cross-product comparisons, as noted by Borusyak, Hull and Jaravel (2022). These approaches would be potentially susceptible to product-level trends, such as technological shocks, being correlated with tariffs, whether incidental or targeted by policymakers.[5] Juhasz (2018) is another approach that also uses a within-country, within-product comparison: she uses the rerouting of international trade routes during the Napoleonic Wars as exogenous variation in protection across French regions. My design has the advantage of not relying on transportation network changes, which may independently affect output and innovation through information flows or market access.

Third, this paper contributes to the trade-and-innovation literature (as reviewed by Shu and Steinwender 2019 and Akcigit and Melitz 2022). My paper finds that the effect of trade protection on innovation depends on where a product stands relative to the technological frontier. In addition, I situate these innovation responses within their broader implications for output growth. My paper is also related to Aghion et al. (2005), who show that theoretically and empirically the relationship between innovation and competition is an inverted U. While the mechanism is closely related — in both frameworks, protection from competition can substitute for innovation as a source of rents — the settings and objects of study differ. Aghion et al. (2005) model competition as the share of profits that neck-and-neck firms can extract through collusion; I instead model protection as tariffs and transportation costs applied to foreign products, which is more specific to the trade setting. In addition, they study how the level of innovation varies with the level of competition, whereas I study how the change in innovation in response to a change in trade protection varies with distance to the

[5] See Appendix B.6 for a formal comparison between the shift-share and the design in this paper.

technological frontier. Aghion et al. (2009) is closer to my paper, but they study how the effect of entry on incumbent innovation is mediated by distance to frontier, whereas I study the effect of trade protection. Amiti and Khandelwal (2013) study how the effect of trade protection on quality upgrading is affected by distance to frontier. Unlike either paper, I situate these innovation responses within their broader implications for output growth.

Fourth, this paper contributes to the industrial policy literature, specifically the ex-post empirical analysis of such policies (as reviewed by Lashkaripour and Wu 2025). My paper addresses numerous identification challenges such as endogenous policy selection and reverse causality as discussed by Juhasz, Lane and Rodrik (2024). In addition, in contrast to other ex-post empirical analyses in this space (e.g. Lane 2025, Juhasz 2018, Choi and Levchenko 2025, Hanlon 2020), my paper focuses on multiple products rather than an individual sector, clarifying a key condition under which trade protection helps or harms.

Roadmap: Section 2 describes the historical context. Section 3 presents a partial equilibrium model of trade and innovation in which distance to the technological frontier shapes the effect of protection. Section 4 develops the empirical framework and identification strategy. Section 5 describes the data. Section 6 presents the results and discusses their interpretation. Section 7 presents a general equilibrium model studying the welfare implications of trade, and how distance to frontier mediates such gains or losses.

2 Historical Context

In the 1850s, the United States was an agrarian economy lagging behind Western Europe's industrial powers — the United Kingdom, France, Belgium, and the German states (which later unified to form Germany). The UK stood furthest ahead, having led the world technologically since the First Industrial Revolution in the 1760s. The US was in the position that many developing countries occupy today: a net exporter of primary products and a net importer of manufactures. In 1850, for example, 53% of steel used domestically was imported despite a large iron ore endowment, while chemicals such as soda ash and chloride of lime were entirely imported with no domestic production of any kind. By the 1890s, the US catches up and becomes a net exporter of manufactures, with GDP per capita 50% higher than at midcentury.

This transformation was accompanied by high tariffs, ranging from 16% to 45% over the second half of the century as Figure 1 shows. This aggregate correlation between tariffs and growth is often cited to justify current protectionist trade policy in the US.[6] My analysis centers on the 1860s, which offer a useful natural experiment: with tariffs supplying 90% of Federal revenues (Irwin 2020), Civil War fiscal pressures drove average rates to their highest level since the Tariff of Abominations in 1828. Thus, this large and sudden tariff increase was plausibly driven by the need for revenue rather than industrial policy goals. Section 4.5 develops this argument in detail.

On the international front, this period is often referred to as the First Globalization, an era of unprecedented economic and financial integration between nations. It was driven by the rise of the steamship and telegraph, adoption of the gold standard, and unilateral and bilateral reduction of protectionist measures. Within the US, exposure to the First Globalization varied greatly depending on geographic distance and the transportation network.[7] As Special Commissioner of the Revenue David A. Wells (1869) put it in a report to Congress,

> "If the duty on pig iron were entirely removed, the American producer in the interior would still enjoy a protection in the cost of transportation ... that intervene between the place of production and the port of entry ... which circumstances renders the transport of a single pound of foreign pig iron to any considerable distance into the interior a matter of ordinary commercial impossibility."

[6] See Donald Trump (2025): "We were at our richest from 1870 to 1913. That's when we were a tariff country" and Oren Cass (2024): "Behind some of the world's highest tariff barriers, the United States transformed from colonial backwater to continent-spanning industrial colossus".

[7] In the modern day, other papers make a similar case. Atkin and Donaldson (2015) show that the probability of finding foreign imports falls the further one goes into the interior. Bosker and Haasbroek (2025) and Cosar and Fajgelbaum (2016) find that access to ports mediates the impact of globalization.

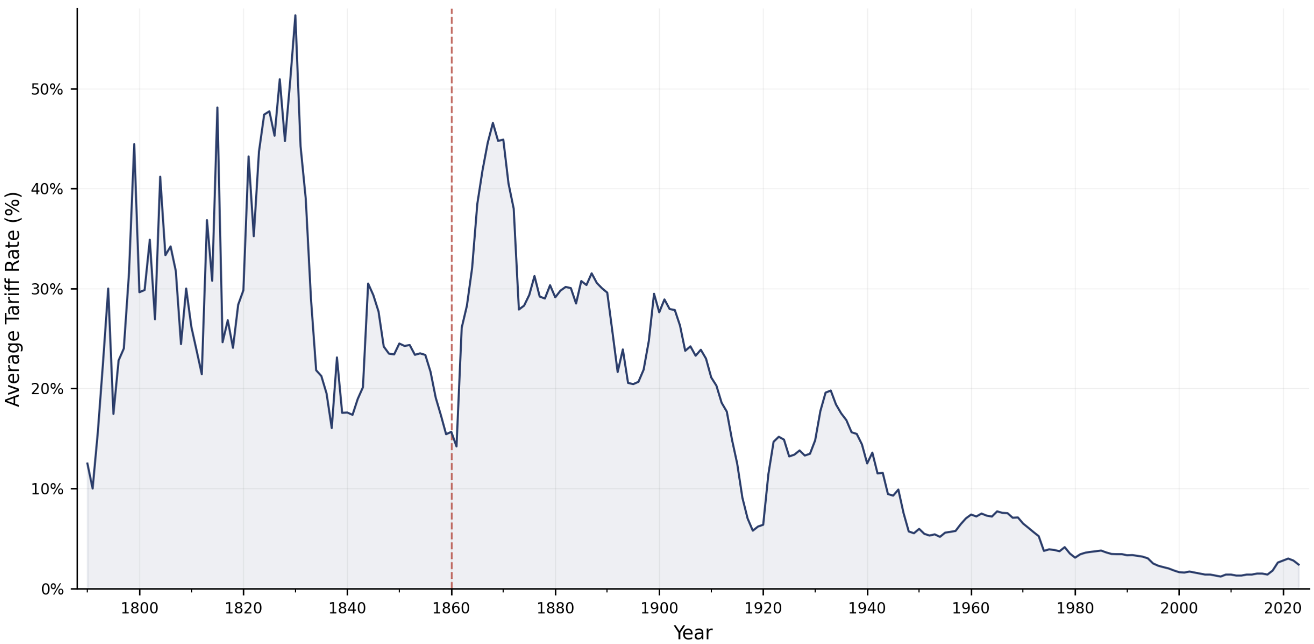

Figure 1: Average US tariff rate on all imports, 1790–2023 (Duties collected as a percentage of total imports for consumption). Data for 1790–2000 come from Table Ee424-430, Irwin (2006). Data for 2001–2023 come from the U.S. International Trade Commission (2024). The dashed vertical line marks 1860.

Trade economist Frank Taussig (1915) shared this view: "The cost of transportation from the sea-board to the interior is such that, even in the absence of the duty, steel rails would be imported only to supply railways near tide-water."[8] High transportation costs were thus a form of protection for interior producers against foreign imports. Because initial levels of protection varied across space, a nationwide tariff shock raises protection proportionally more in coastal counties than in the interior – a feature I exploit in my empirical design in Section 4.1.[9]

An attractive feature of this period is the lack of a retaliatory response by trading partners. In Appendix A.1, I discuss the reasons why the European powers did not retaliate against the US for its high tariffs. This is useful because it isolates the import protection

[8] See also Sundararajan (1970): "... in the vast hinterland of the United States foreign competition is checked due to the existence of transportation costs. Whatever handicaps to American producers may be imposed by those costs, the influence of transport costs on competition from foreigners is felt only at the seacoast and adjacent areas."

[9] Gravity models in trade also imply the output response to a nationwide tariff shock is larger for coastal than interior areas. See Appendix B.5 for a formal analysis.

channel from retaliatory tariffs and loss of export markets.

Another important dimension of this period is the wide variation in American patenting activity across products, which I use to proxy for technological capabilities. As Table 1 shows, in some industries, such as cotton textiles, American producers were innovating at rates similar to Britain's – the result of decades of transatlantic technology transfer dating back to the early republic (Jeremy 1981). In others, such as paints, the US lagged far behind: Sherwin-Williams was the first US paint firm to employ a chemist in 1884, over a century after British rivals such as Berger Paints (1760) and Winsor & Newton (1832) were founded by chemists.[10] As I show in this paper, this heterogeneity in technological proximity has sharp implications for how protection affects innovation incentives and output growth.

Table 1: Distance to Technological Frontier: British and US Patents, 1855-1859

Product	Distance to Frontier	British Patents	US Patents
Cotton Textiles	1.31	464	353
Worsted Goods	1.31	464	353
Boots	1.71	181	106
Hats	1.86	169	91
Iron Bar	2.53	374	148
Iron Pig	2.53	374	148
Steel	2.53	374	148
Paper	2.79	301	108
Beer	2.87	149	52
Whiskey	2.87	149	52
Glass	3.39	95	28
Gloves	3.64	222	61
Rope	3.88	128	33
Nails	4.98	214	43
Paints	7.82	297	38

Notes: Distance to frontier is the ratio of British to American patents in 1855–1859. A value of one indicates technological parity; higher values indicate the US lags further behind. Patent data comes from Hanlon (2016) and Shaw (undated, accessed 2025). See Section 5.1 for details on the construction of this data.

[10] See Appendix A.2 for more examples and validation of my distance to frontier measure.

3 A Model of Protection and Distance to Frontier

In this section, I present a model of trade and innovation in which distance to the technological frontier shapes the effect of protection on output. The model introduces the main objects and frames the empirical analysis in the sections that follow. Its key result is that trade protection can reduce output in partial equilibrium – a prediction absent from gravity models. The crucial mechanism operates via variable markups: for sufficiently close-to-frontier products, the domestic producer already commands a large market share and faces sufficiently inelastic residual demand that protection reduces the incremental gain from innovation. If this innovation loss is large enough, it can overtake the positive substitution effect — the shift in demand from foreign to domestic varieties that standard models predict — and reduce output on net.

The model also provides an imperfect competition rationale for infant industry protection distinct from the usual external-economies-of-scale or learning-by-doing arguments. For products sufficiently far behind the frontier, protection makes the market contestable enough that the domestic producer can capture the returns from innovating, so the innovation effect amplifies the positive substitution effect. In this case innovation and protection are complements rather than substitutes. Later in Section 7, I study the welfare effects of trade and how distance to frontier shapes it.

Sections 3.1 and 3.2 set up the model mathematically. Sections 3.3 and 3.4 derive comparative statics and provide the economic mechanism for why protection has different effects across the frontier spectrum.

3.1 Setup

In county i's market for product z, a local producer competes á la Bertrand against a UK exporter. Local expenditure on product z is E_{iz}, which is held fixed in this partial-equilibrium model. In Section 7, I relax this assumption and allow expenditures E_{iz} to adjust in general equilibrium.

Demand is CES with elasticity $\sigma > 1$. If the local and UK prices are p_{iz}^{US} and p_{iz}^{UK} respectively, demand for the local variety is:

$$q_{iz}^{US} = (p_{iz}^{US})^{-\sigma} P_{iz}^{\sigma-1} E_{iz}$$

where the price index is $P_{iz} = \left[(p_{iz}^{US})^{1-\sigma} + (p_{iz}^{UK})^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$. The share of the local market i captured by the local firm is:

$$\lambda_{iz} = \frac{(p_{iz}^{US})^{1-\sigma}}{(p_{iz}^{US})^{1-\sigma} + (p_{iz}^{UK})^{1-\sigma}}.$$

The (quality-adjusted) productivity of the UK is T_z^{UK}. I assume all counties in the US share the same initial technology T_z^{US}. Define distance to the frontier as:

$$D_z := \frac{T_z^{UK}}{T_z^{US}}.$$

Thus D_z close to one means the US is close to the British frontier, while larger D_z means the US is further behind. Unit costs for the UK are:

$$c_z^{UK} = \frac{1}{T_z^{UK}}$$

Let t_z denote the ad valorem tariff on product z, and let τ_{iz} be the freight cost of delivering the product to county i. Total protection is thus:

$$TP_{iz} := t_z + \frac{\tau_{iz}}{c_z^{UK}}$$

Interior regions have higher τ_{iz} and thus higher total protection TP_{iz}. The local firm can pay a fixed cost F to upgrade its technology from T_z^{US} to αT_z^{US}, with step size $\alpha > 1$.[11] In

[11] In Appendix D.5, I study a continuous-innovation version of this model, which delivers additional comparative statics results and maps more closely to the empirical specification for patents.

this section, I am agnostic to whether this upgrade reflects domestic innovative activity or adoption of existing technology, but I refer to this upgrading as innovation throughout as a shorthand. Let $I_{iz} \in \{0, 1\}$ denote the local firm's decision to innovate. Thus, the local firm's unit cost is:

$$c_{iz}^{US} = \frac{1}{T_z^{US} \alpha^{I_{iz}}}$$

3.2 Pricing and Innovation

Firms compete in prices a la Bertrand. Under CES demand, the perceived own-price elasticity of the local firm is

$$\varepsilon_{iz}^{US} = \sigma - (\sigma - 1)\lambda_{iz}.$$

Therefore the Bertrand first-order conditions imply

$$p_{iz}^{US} = \frac{\varepsilon_{iz}^{US}}{\varepsilon_{iz}^{US} - 1} c_{iz}^{US} = \frac{\sigma - (\sigma - 1)\lambda_{iz}}{(\sigma - 1)(1 - \lambda_{iz})} c_{iz}^{US},$$

and

$$p_{iz}^{UK} = \frac{1 + (\sigma - 1)\lambda_{iz}}{(\sigma - 1)\lambda_{iz}} (1 + TP_{iz}) c_z^{UK}.$$

Let

$$r_{iz} \equiv \frac{c_z^{UK}(1 + TP_{iz})}{c_{iz}^{US}} = \frac{\alpha^{I_{iz}}(1 + TP_{iz})}{D_z}$$

denote relative local competitiveness. Combining the CES share equation with the Bertrand pricing equations, the local expenditure share $\lambda_{iz}(r_{iz})$ is the solution to the following implicit equation:

$$\frac{\lambda_{iz}(r_{iz})}{1 - \lambda_{iz}(r_{iz})} = \left[\frac{[1 + (\sigma - 1)\lambda_{iz}(r_{iz})][1 - \lambda_{iz}(r_{iz})]}{\lambda_{iz}(r_{iz})[\sigma - (\sigma - 1)\lambda_{iz}(r_{iz})]} r_{iz} \right]^{\sigma - 1}.$$

Operating profit is sales divided by the perceived elasticity:

$$\Pi_{iz}^{op} = E_{iz}\frac{\lambda_{iz}}{\varepsilon_{iz}} = E_{iz}\frac{\lambda_{iz}}{\sigma - (\sigma-1)\lambda_{iz}}$$

If $I_{iz} = 0$, the local firm has relative competitiveness $(1+TP_{iz})/D_z$. If $I_{iz} = 1$, the local firm has relative competitiveness $\alpha(1+TP_{iz})/D_z$. The gain from innovation is the resulting profit differential:

$$G_{iz}(D_z, TP_{iz}) := \Pi_{iz}^{op}\left(\frac{\alpha(1+TP_{iz})}{D_z}\right) - \Pi_{iz}^{op}\left(\frac{(1+TP_{iz})}{D_z}\right)$$

The firm innovates if this gain exceeds the fixed cost:

$$I_{iz}^{*} = 1\{G_{iz}(D_z, TP_{iz}) \geq F\}.$$

Local output is:

$$Y_{iz} = E_{iz}\lambda_{iz}\left(\frac{\alpha^{I_{iz}^{*}}(1+TP_{iz})}{D_z}\right).$$

3.3 Comparative Statics

Let TP_{iz}^0 and $TP_{iz}^1 > TP_{iz}^0$ denote pre- and post-protection levels. We can decompose the change in output into two components:

$$\begin{aligned}\Delta \log Y_{iz} = &\underbrace{\log Y_{iz}(I_{iz}^{*}(TP_{iz}^0), TP_{iz}^1) - \log Y_{iz}(I_{iz}^{*}(TP_{iz}^0), TP_{iz}^0)}_{\text{substitution effect } > \ 0} \\ &+ \underbrace{\log Y_{iz}(I_{iz}^{*}(TP_{iz}^1), TP_{iz}^1) - \log Y_{iz}(I_{iz}^{*}(TP_{iz}^0), TP_{iz}^1)}_{\text{innovation effect}}.\end{aligned} \tag{1}$$

The first term is the substitution effect. It holds the optimal innovation decision fixed at its pre-protection value, $I_{iz}^{*}(TP_{iz}^0)$, and measures how output changes when protection rises from TP_{iz}^0 to TP_{iz}^1. The second term is the innovation effect. It holds protection fixed at its post-change level, TP^1_{iz}, and measures the additional change in output caused by the induced change in the innovation decision from $I^*_{iz}(TP^0_{iz})$ to $I^*_{iz}(TP^1_{iz})$.[12]

The substitution effect is always positive: an increase in protection raises the relative price of the foreign substitute and induces local consumers to substitute toward the domestic variety in partial equilibrium, i.e. holding fixed total expenditures.

The innovation effect captures the additional change in output induced by the change in innovation. Holding protection fixed, innovation raises domestic productivity, lowers marginal cost, and increases competitiveness, so the output gain from moving from $I_{iz} = 0$ to $I_{iz} = 1$ is positive. Therefore, the sign of the innovation effect depends on whether protection raises or lowers innovation, i.e. the sign of $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})}$.

Under Bertrand oligopoly, $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})}$ can be positive or negative. For a near-frontier firm that already commands a large market share, the firm is on a sufficiently inelastic portion of its residual demand curve that the incremental gain from innovation can fall. Thus, in some cases, protection makes it profit-maximizing to save on innovation costs, so $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})} < 0$. If this innovation loss is large enough, it can overtake the substitution effect, resulting in a net negative effect of protection on output.

This possibility of a negative innovation reaction relies crucially on variable markups. Under perfect competition, there are no economic profits, so there is no profit-based motive to innovate: $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})} = 0$. The innovation channel is absent entirely. Under monopolistic competition with CES demand, each firm faces a constant markup of $\frac{\sigma}{\sigma-1}$. As the firm is too small to affect the competitive environment and the elasticity of demand is constant, the gain from innovation is a fixed fraction of pre-innovation profits, so protection always raises the incentive to innovate, i.e. $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})} > 0$.

[12] The decomposition can instead be written by first allowing innovation to adjust and then changing protection, rather than first changing protection then allowing innovation to adjust. The main insights are unchanged.

3.4 Main Propositions

I now state the main comparative statics results formally and provide proofs.

Proposition 1: There always exists a threshold $D_{iz}^{I,*}$ such that for all $D_z \geq D_{iz}^{I,*}$, protection weakly increases innovation:

$$\frac{\Delta I_{iz}^*}{\Delta \log(TP_{iz})} \geq 0$$

Below this threshold, protection can reduce innovation.

Proof. See Appendix D.3.1.

Proposition 2: There always exists a threshold $D_{iz}^{Y,*}$ such that for all $D_z \geq D_{iz}^{Y,*}$, protection increases output:

$$\frac{\Delta \log Y_{iz}}{\Delta \log(TP_{iz})} > 0.$$

Below this threshold, protection can reduce output.

Proof. See Appendix D.3.2.

Appendix D.1 illustrates the possibility of protection causing innovation and output to fall with a numerical example, with $\sigma = 8$, $\alpha = 3$, $D_z = 1.05$, and a 10 percentage point increase in protection. By contrast, gravity models predict that unilateral protection — absent any retaliatory response — weakly increases output in partial equilibrium (see Appendix B.5). With further assumptions, we can obtain a stronger single-crossing condition, where below the threshold, a negative response of output and innovation is guaranteed.

Corollary 1: Innovation Single-Crossing From Below. Suppose $G_{iz}(D_z, TP_{iz})$ is single-peaked in D_z and $G_{iz}(1, TP_{iz}^0) \geq F > G_{iz}(1, TP_{iz}^1)$. Then on the domain $D_z \geq 1$, protection strictly reduces innovation for all $D_z < D_{iz}^{I,*}$ and weakly increases it for all $D_z \geq D_{iz}^{I,*}$:

$$\frac{\Delta I_{iz}^*}{\Delta \log(TP_{iz})} < 0 \quad \text{for } D_z < D_{iz}^{I,*}, \qquad \frac{\Delta I_{iz}^*}{\Delta \log(TP_{iz})} \geq 0 \quad \text{for } D_z \geq D_{iz}^{I,*}.$$

Proof. See Appendix D.3.3.

Corollary 2: Output Single-Crossing From Below. Suppose in addition that $\alpha > (1 + TP^1_{iz})/(1 + TP^0_{iz})$. Then on the domain $D_z \geq 1$, protection strictly reduces output for all $D_z < D^{Y,*}_{iz}$ and strictly increases it for all $D_z \geq D^{Y,*}_{iz}$:

$$\frac{\Delta \log Y_{iz}}{\Delta \log(TP_{iz})} < 0 \quad \text{for } D_z < D^{Y,*}_{iz}, \qquad \frac{\Delta \log Y_{iz}}{\Delta \log(TP_{iz})} > 0 \quad \text{for } D_z \geq D^{Y,*}_{iz}.$$

Proof. See Appendix D.3.4.

Taken together, Propositions 1 and 2 say that protection increases output and innovation for products sufficiently far from the frontier. The novel result is that protection can potentially reduce output and innovation for products close to the frontier. Whether this negative region is operative is an empirical matter, dependent on the setting, policy shock and products in question. Corollaries 1 and 2 provide a single-crossing property, ensuring that products below the threshold see a decrease in output and innovation.

These theoretical results motivate the empirical linear specification in Equation (3), which interacts distance to frontier with trade protection so that the effect of protection is allowed to change sign across products. I structurally interpret the regression coefficients through the lens of this model in Section 6.4.

Note that the heterogeneity in responses across products arises from variable markups interacting with differences in initial technology levels. Additional heterogeneity on the cost side, such as larger innovation step sizes for more technologically backward products, i.e. $\frac{\partial \log \alpha_z(D_z)}{\partial D_z} > 0$ — the "advantage of backwardness" of Gerschenkron (1962) — would further amplify these forces. Similarly, learning-by-doing externalities would amplify the gradient, as far-from-frontier products see larger output increases under protection and are earlier on the learning curve.

This simple model has a number of additional interesting implications. One is that with endogenous innovation, trade protection can actually lead to markups falling.[13] Another implication is that markups can positively or negatively covary with innovation. Therefore, rising markups are consistent with innovation increasing, remaining constant, or falling, and cannot on their own be taken as evidence of declining innovative activity.

4 Empirical Framework

This section develops the empirical framework. I first discuss the key measure of protection that incorporates both tariffs and transportation costs (Section 4.1). I then present the econometric specification (Section 4.2) and instrument (Section 4.3), followed by identification arguments (Section 4.4–4.6).

4.1 Treatment

Recall from Section 3 that total protection is the tariff t_z on product z and the freight cost τ_{iz} of delivering the foreign good to county i. Empirically, I measure it as

$$TP_{iz} := t_z + \frac{\tau_{iz}}{p_z} \tag{2}$$

where p_z is the price of the British good at a US port, the empirical counterpart to unobserved British marginal costs c_z^{UK} in the theory section. To economize on notation, I write tariffs in ad valorem form throughout. The empirics take into account whether a tariff line is specific, ad valorem or composite; Appendix B.4 presents more general formulas including specific tariffs. The main intuitions are unchanged.

If $\tau_{iz} = 0$, (i.e. no transportation costs), then TP_{iz} simply reduces to the tariff t_z, which

[13] Without endogenous innovation, protection always raises markups: the tariff increases the delivered cost of foreign goods, raising the domestic firm's market share and thus its markup under Bertrand competition. With endogenous innovation, however, protection can eliminate the incentive to innovate (Proposition 1), reducing the domestic firm's competitiveness from αT_z^{US} to T_z^{US} and increasing marginal cost. If marginal costs rise more than prices, markups fall. It can be shown this case holds in the scenario where the innovation effect is negative and overtakes the mechanical substitution effect, as the firm faces a more elastic residual demand curve, limiting the extent of its price increase.

is the notion of protection used in the majority of empirical trade papers (e.g. Pavcnik 2002, Bustos 2011, Khandelwal and Topalova 2011, De Loecker et al. 2016, Fajgelbaum et al. 2020).

For $\tau_{iz} > 0$, transportation costs are an additional form of protection as they serve to increase delivered import prices much like a tariff would, as in Juhasz (2018) and Pascali (2017). As discussed in Section 2, contemporary observers such as Wells (1869) and Taussig (1915) argued that transportation costs rendered foreign imports commercially unviable in the interior — effectively functioning as a tariff. See Figure 2 for a visual depiction of transportation costs across space for the product pig iron. Coastal counties near major ports such as New York and Philadelphia face negligible transportation costs relative to import prices. By contrast, for interior Midwest counties, transportation costs alone exceed 45% of the import price — roughly twice the protection provided by the 24% pig iron tariff in place in 1860. In this sense, transportation costs and tariffs are two sides of the same coin: they both serve to increase delivered import prices for domestic consumers, thereby inducing substitution from foreign to local varieties.[14]

I leave the specification of τ_{iz} relatively general, allowing for markup pricing by the transportation sector, and discipline it empirically in Section 5.2 using newly digitized commodity-specific freight rates. As that section shows, transportation costs are mostly additive, contrary to the "iceberg" assumption commonly made in trade and spatial. Thus, high transportation costs offer more protection for cheaper (low p_z) products rather than expensive ones, as the transportation cost will be a larger fraction of the price for cheaper goods. Correctly specifying protection levels therefore requires port prices by product, which I digitize alongside the tariff data.

Protection, in addition to making competitor import products more expensive, also makes foreign inputs used by domestic firms more expensive. Thus, I define an input protection measure in the following way:

[14] One important difference is that tariffs generate revenues for the Federal Government. This distinction becomes important in the welfare analysis in Section 7.

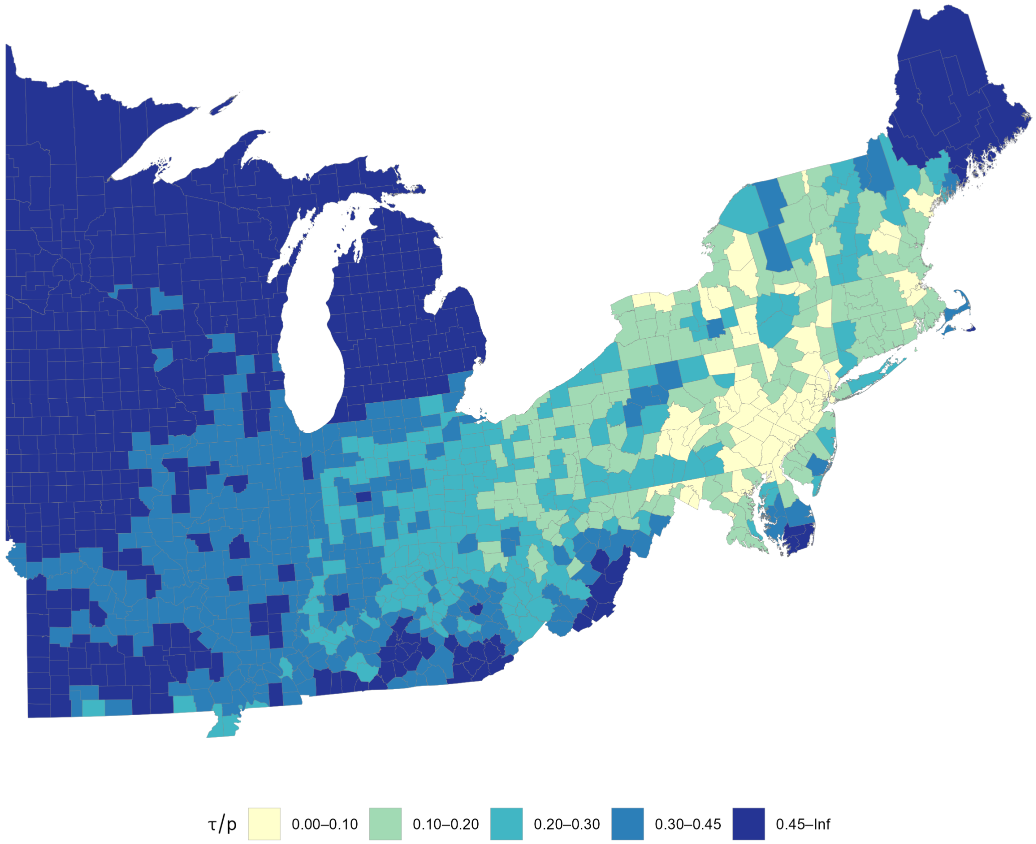

Figure 2: Transportation costs as a fraction of import price at port (τ_{iz}/p_z) for pig iron, 1860. Darker shading indicates higher transportation costs relative to port price, implying greater natural protection from foreign competition.

$$\log(Input\_TP_{iz}) = \sum_{z'} \zeta_{z' \to z} \log(TP_{iz'})$$

where $\zeta_{z' \to z}$ is the Leontief share of input z' used in the production of z, using the state-level census of manufactures digitized by Atack, Bateman and Weiss (2006). See Section 5.1 for details on the construction of $\zeta_{z' \to z}$. I include this variable in all specifications.

4.2 Econometric Specification

I estimate the following regression, expressed in growth rates:

$$\Delta \log y_{iz} = \beta[\Delta \log(TP_{iz}) \times Dist\_Front_z] + \theta \Delta \log(TP_{iz}) + \gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz} \quad (3)$$

where Δ is the difference operator (i.e. $\Delta \log y_{iz} = \log y_{iz,1870} - \log y_{iz,1860}$), i, z index county and product respectively. $\Delta\delta_i$ is a county fixed effect, $\Delta\delta_z$ is a product fixed effect and X_{iz} is a vector of controls, including the input protection measure defined in Section 4.1. θ is the baseline effect of protection for a product with distance to frontier of zero, and β captures how the effect of protection varies with distance to the frontier. The interaction of distance to frontier with protection is motivated by the model in Section 3, allowing for a potential sign flip in the effect of protection on output.

First, note that taking the difference controls for any persistent county-product factors δ_{iz}. For example, this would control for Pittsburgh being better at producing steel than Boston. Second, the county fixed effect $\Delta\delta_i$ controls for local equilibrium effects such as agglomeration, labor migration and local consumer price indices.[15] This would, for example, control for Chicago growing rapidly in all product categories. Third, unlike other designs, I am also able to include product fixed effects $\Delta\delta_z$, controlling for any nationwide trends in products such as technological developments. Thus, this design isolates within-product, across-county changes in trade protection.

In words, this design is comparing the growth rate of Philadelphia steel to Chicago steel when the change in protection is higher for Philadelphia steel than Chicago steel (while controlling for the average growth rate in Philadelphia and Chicago across all products). If Philadelphia steel grows faster than Chicago steel, we infer that output protection benefits growth for steel. One notable feature of this design is that identification operates entirely

[15]This fixed effect makes identification credible, but then we must interpret the parameter β as the effect of protection on output, **holding** real incomes fixed. When protection increases, real incomes can decrease due to higher import goods costs, or increase due to increased demand. In other words, this is a partial equilibrium elasticity rather than a general equilibrium elasticity. See also discussion in Chodorow-Reich (2019, 2020) and Guren et al. (2021).

within-product — comparing counties producing the same product — rather than across products, as is common in the empirical trade literature.

However, the error term Δe_{iz} may still be correlated with $\Delta \log(TP_{iz})$. For example, newly built railroads may facilitate access to domestic markets and inputs, in addition to affecting protection from foreign imports. This would lead to biased estimates for β and θ. Thus I turn to an instrument to deal with this concern.

4.3 Instrument

Because changes in transportation costs may affect output through other channels – via access to inputs or markets for example – I instrument for $\Delta \log(TP_{iz})$ using only the tariff-driven component of protection changes. To do so, I use a first order approximation (see Appendix B.1 for derivation):

$$\Delta \log(TP_{iz}) \approx \underbrace{STS^t_{iz}\Delta \log t_z}_{IV} + STS^\tau_{iz}\Delta \log \tau_{iz} - STS^\tau_{iz}\Delta \log p_z \tag{4}$$

where $STS^t_{iz} = \frac{t_z p_z}{t_z p_z + \tau_{iz}}$ and $STS^\tau_{iz} = \frac{\tau_{iz}}{t_z p_z + \tau_{iz}}$.

STS^t_{iz} captures the tariff's share in total protection for a given county-product iz pair. This expression has a "shift-exposure" structure as in Borusyak and Hull (2023), with cross-sectional exposures to aggregate shifters.[16]

I use the first component $STS^t_{iz}\Delta \log t_z$ as an instrument. Intuitively, counties where the tariff forms a larger fraction of local protection (i.e. those with high STS^t_{iz}) see larger percentage increases in protection for the same nationwide tariff increase. As coastal areas will generally have larger values of STS^t_{iz} than interior regions, this formalizes the intuition from Section 2 that tariffs have larger effects for coastal areas than interior areas.

[16]This is technically not a shift-share according to the definition in Borusyak, Hull and Jaravel (2025), though in spirit it resembles the same structure of a local exposure to an aggregate shifter. Borusyak and Hull (2023) show that shift-shares are a special case of the "shift-exposure" framework.

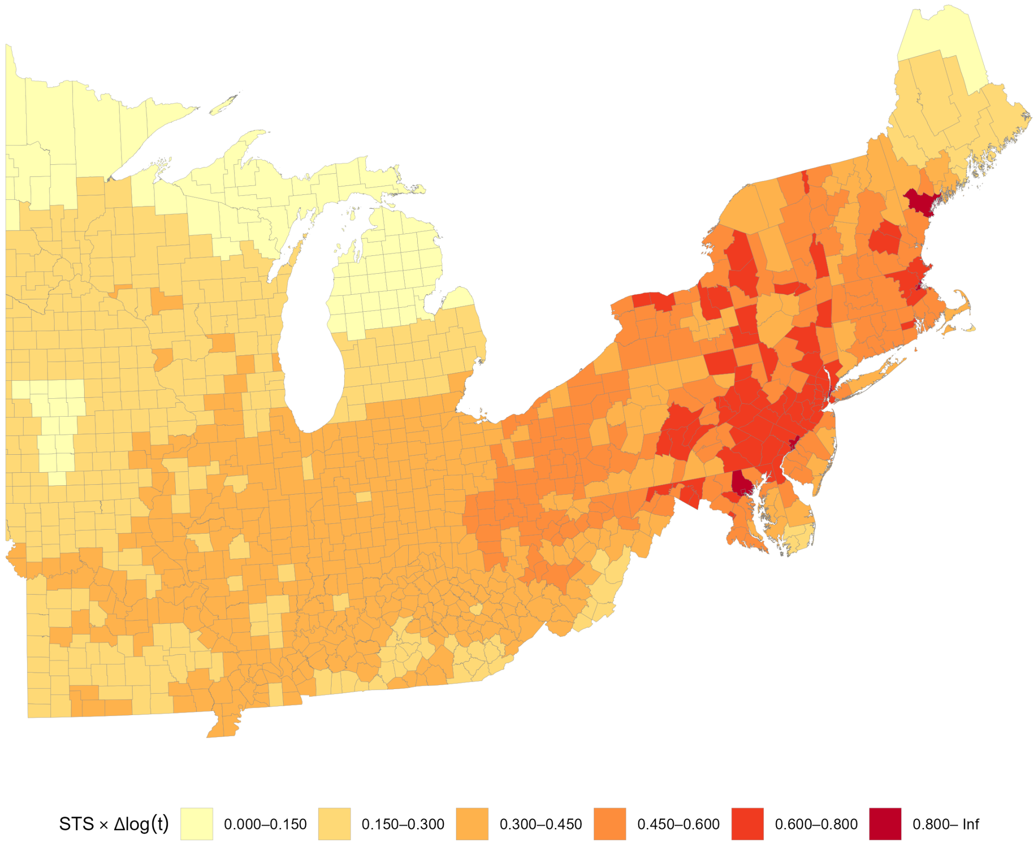

Figure 3: Instrumental variable ($STS_{iz}^{t} \times \Delta \log t_z$) for pig iron, 1860–1870, as defined in Equation (4). Darker shading indicates greater exposure to the nationwide tariff shock, with coastal counties near major ports receiving the largest percentage increases in protection.

One way to think about this is to note that interior regions have two forms of protection – tariffs and internal transportation costs – so a given tariff increase raises their protection by a smaller proportion than in coastal areas, which rely on the tariff alone. See Figure 3 for a visual depiction of the tariff-driven component of protection changes across space for pig iron. Coastal counties near major ports such as New York and Philadelphia see the largest percentage increases in protection, while interior counties — already shielded by high freight costs — see much smaller changes despite facing the same nationwide tariff increase.

The instrument developed in this section isolates the change in protection arising only from the change in tariffs.[17] The next three subsections will then argue that this component is exogenous to county-product shocks.

4.4 Identification

Formally, we need the following IV orthogonality condition to hold for identification of β and θ:

$$\mathbb{E}(STS_{iz}^{t}\Delta\log t_z \cdot \Delta e_{iz}|\Delta\delta_i, \Delta\delta_z) = 0 \tag{5}$$

A sufficient condition for Equation (5) is:

$$\mathbb{E}(\Delta\log t_z \cdot \Delta e_{iz}|\Delta\delta_i, \Delta\delta_z, STS_{iz}^{t}) = 0 \tag{6}$$

See Appendix B.2 for the proof. Condition (6) requires **nationwide** tariff changes to be uncorrelated with **local** product shocks, conditional on product and county fixed effects.

It is instructive to compare this condition with conditions required by other designs. One canonical approach is to take a nationwide trade liberalization episode and compare the change in outcomes for sectors that received larger tariff cuts to sectors that received smaller cuts (such as in Pavcnik (2002), Bustos (2011), De Loecker et al. (2016) and Flaaen and Pierce (2024)). For these cross-product comparisons, the equivalent condition would be: $\mathbb{E}(\Delta\log t_z \cdot \Delta e_z|\Delta\delta) = 0$, which requires **nationwide** tariff shocks to be uncorrelated with **nationwide** – as opposed to **local** – product shocks. Thus, if policymakers are targeting nationwide product trends, such as technological breakthroughs[18], this would potentially confound a nationwide cross-product empirical design, but not my design due to the presence of the product fixed effect $\Delta\delta_z$.

In addition, as Borusyak, Hull and Jaravel (2022) show, shift-share designs based on

[17] Another way to think about this is that this is the change in protection if the US had no new railways built in the decade 1860-1870 and port prices did not change.

[18] For example, the Bessemer process, invented in 1856, enabled the mass production of steel for the first time in human history.

exogenous shocks – such as the China Shock paper of Autor, Dorn and Hanson (2013) – rely on a similar identification argument as nationwide cross-product studies. Thus shift-shares based on the exogenous shocks view are also potentially confounded if products receiving different tariffs are on differential trends, while my approach is robust to this concern.[19] See Appendix B.6 for a detailed comparison of this design with the shift-share design.

There are two major threats to Condition (6) holding: 1) endogenous tariff setting by policymakers, and 2) other shocks that are coincidentally correlated with tariffs. The rest of the section will discuss the trade policy environment, provide a narrative for why neither endogeneity concern is likely to hold in my context, and present a balance test showing that county-product pairs receiving more protection are comparable on observable characteristics to those receiving less.

4.5 Policy Environment and Narrative

In this period, 90% of Federal revenues came from tariffs (Irwin 2020), and federal budgets and surpluses played a pivotal role in tariff policy. In November 1860, Abraham Lincoln is elected President. Out of fears that Lincoln would restrict slavery, Southern states begin seceding from the Union and the Civil War begins two months after Lincoln's inauguration. While many assumed the War would end by Christmas 1861, it drags on and turns out to be an unprecedented fiscal burden, an order of magnitude larger than any other fiscal shock in US history up until that point. Figure 4 shows the dramatic spike in federal expenditures and deficits during the war years.

To finance the war effort, Congress passes five major tariff increases (and many more minor revisions) in 1861-1865. For comparison, Congress only passed one major tariff bill in the 1850s. Tariffs went from 16% in 1860 to 45% by 1870. It seems clear that revenue generation was policymakers' most urgent concern. The logic of revenue maximization implies

[19] The other path to identification for shift-shares, based on the exogeneity of shares (Goldsmith-Pinkham et al. 2020), is unlikely to hold in my context. Locations specializing in different sectors are likely to be exposed to a variety of different shocks such as immigrant flows, technological changes and unionization drives.

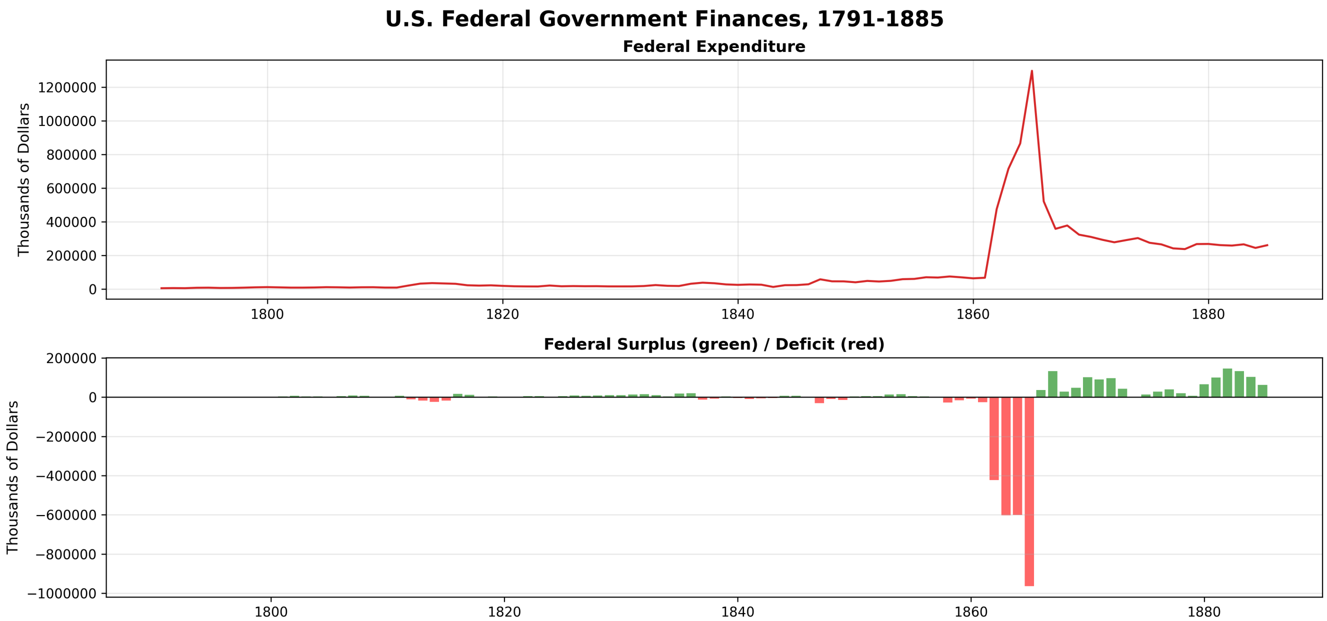

Figure 4: U.S. Federal Government Finances, 1791–1885. Top panel: federal expenditure in thousands of 1870 dollars. Bottom panel: federal surplus (green) and deficit (red) in thousands of 1870 dollars. The spike in expenditures and deficits during the 1861–1865 period reflects Civil War fiscal pressures. Data from Carter et al. (2006), Historical Statistics of the United States, Millennial Edition.

targeting products with low elasticities of demand, a national product characteristic, not a local one. The same holds for minimizing deadweight loss subject to a revenue target, as in the classic Ramsey (1927) rule. Such targeting is absorbed by the product fixed effect $\Delta\delta_z$ and poses no threat to Condition (6). Nonetheless, other goals – such as industrial development or protecting a lobby – could have been pursued in tandem. A key advantage of my study is that it is robust to any product-level trend correlated with tariff changes, whether targeted by policymakers or merely coincidental. This is not true of other approaches, such as shift-shares or cross-sector analyses.

In order for endogenous tariff setting to pose a threat to Condition (6), policymakers would need to be reacting to idiosyncratic county-product shocks in a way that systematically deviates from the county-time, county-product and product-time average. First, policymakers would need to identify county-product trends that deviate from the product-time, county-time and county-product averages – across roughly 1,500 counties and hundreds of tariff line items. Then they would need to set tariff levels to systematically exploit these residual trends. Finally, they would need to build majority consensus on this targeting in both the House and Senate. This is already a high cognitive and informational burden on 19th-century policymakers in peacetime – one only exacerbated by the wartime fiscal pressures. In contrast, targeting national product-level trends is more tractable and easier to build Congressional consensus around.

This assuages concerns about endogenous tariff setting, but other Civil War shocks – such as military procurement, Emancipation and the effect of battles – loom large. I address these concerns in Section 6.2.

4.6 Balance Tests

While Condition (6) cannot be directly tested, balance tests provide suggestive evidence. If county-product pairs receiving more protection differ systematically on observable characteristics from those receiving less, it would suggest the threat of endogenous tariff setting or confounding shocks looms large—either of which would violate Condition (6). For example, systematic imbalance might indicate that policymakers were able to overcome the informational constraints discussed above and selectively target county-product pairs with certain characteristics. This would then raise concerns they could also identify and target idiosyncratic county-product trends, violating Condition (6).

I regress a number of initial period variables on the change in input and output protection, with product and county fixed effects.[20] These characteristics include total labor (county-product pairs with larger labor forces may have more electoral and lobbying power), firm size (county-product pairs with larger firms may be able to overcome collective action problems and lobby as in Olson (1965)) and output (implying more resources for lobbying).

[20]Mathematically:

$$\log X_{iz,1860} = \phi \left(STS^t_{iz} \times \Delta \log t_z \right) + \psi \sum_{z'} \zeta_{z' \to z} \left(STS^t_{iz'} \times \Delta \log t_{z'} \right) + \delta_z + \delta_i + \epsilon_{iz}$$

Table 2: Balance Test

	IV Output		IV Input	
Dependent Variable	Coef.	SE	Coef.	SE
Log Total Labor Initial	−1.921	(2.854)	−3.299	(2.051)
Log Establishments Initial	−2.015	(1.498)	−0.739	(1.098)
Log Capital Initial	−0.286	(2.369)	−2.545	(2.418)
Log Labor Productivity Initial	−0.694	(1.489)	0.539	(0.540)
Log Output Initial	−2.194	(1.908)	−3.190	(1.832)
Log Output per Estab. Initial	−0.183	(1.929)	−2.405	(1.496)
Log Average Wage Initial	−0.635	(0.443)	0.242	(0.357)
Log Capital-Labor Ratio Initial	1.635	(2.696)	0.754	(1.094)

Notes: Each row reports coefficients from a regression of the baseline (1860) characteristic on changes in the IVs for output and input protection, with county-year and product-year fixed effects. See footnote 20 for the econometric specification. Standard errors clustered at the county and product level in parentheses. None of the 16 coefficients are statistically significant at conventional levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 2 presents the results. None of the sixteen coefficients is significant at the 10% level: county-product pairs receiving larger increases in protection are balanced on initial period observable characteristics against those receiving smaller increases, assuaging concerns of selective targeting by policymakers.[21]

The value of the within-product comparison becomes clear when I run the same balance test, but omit the product fixed effect δ_z, which controls for any nationwide product trend common to all units, such as technological breakthroughs. As we see in Table 3, the number of significant correlations goes from 0/16 to 6/16 (at the 10% level), raising concerns that Condition (6) does not hold when we also exploit cross-product variation. It appears output protection is skewed towards county-product pairs with higher initial output, higher wages and higher labor productivity. Isolating the within-product comparison is thus crucial for the exogeneity narrative — and is infeasible in canonical approaches, including both shift-share

[21] While statistically insignificant, the magnitudes of the output IV coefficients for labor, establishments and output are economically meaningful. Results are robust to including these variables as controls in the main specification. In addition, the sign runs opposite to what a lobbying story would predict: county-product pairs with higher initial output have more resources with which to lobby the central government, and so would be expected to receive more protection, not less.

and nationwide liberalization designs.

Table 3: Imbalance Test: Without Product-Year Fixed Effects

	IV Output		IV Input	
Dependent Variable	Coef.	SE	Coef.	SE
Log Total Labor Initial	−0.143	(0.151)	−0.885*	(0.444)
Log Establishments Initial	0.018	(0.056)	−0.232	(0.170)
Log Capital Initial	0.032	(0.128)	−0.877*	(0.437)
Log Labor Productivity Initial	0.264***	(0.089)	0.331	(0.197)
Log Output Initial	0.208*	(0.107)	−0.430	(0.355)
Log Output per Estab. Initial	0.189	(0.131)	−0.192	(0.397)
Log Average Wage Initial	0.062**	(0.024)	0.192**	(0.081)
Log Capital-Labor Ratio Initial	0.175	(0.124)	0.008	(0.272)

Notes: Each row reports coefficients from a regression of the baseline (1860) characteristic on changes in the IVs for output and input protection, with county-year fixed effects only (no product-year fixed effects). The econometric specification is identical to the one in footnote 20, but without the product fixed effect δ_z. Standard errors clustered at the county and product level in parentheses. 6 of 16 coefficients are statistically significant at the 10% level, in contrast to 0 of 16 when product-year fixed effects are included (Table 2). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5 Data

5.1 Data

I digitize US tariffs and foreign import prices entering US ports from the Foreign Commerce and Navigation Tables of the United States for 1859 and 1869, and the Report of the Secretary of the Treasury on the State of the Finances for 1859. These tables record, for each product, the total quantity imported and total customs value, from which I infer per-unit import prices (see Figure 5 for an example). They also list the applicable tariff — whether ad valorem, specific, or composite — for each tariff line item.

As there is no input-output table available in this period, I construct one using Atack, Bateman and Weiss' (2006) digitized state-level census of manufactures samples, which records input-specific usage by product.[22]

[22]I limit the sample to single product establishments, as, like in the vast majority of manufacturing

No 9.—*General statement of foreign imports*—Continued.

	WHENCE IMPORTED.	MERCHANDISE PAYING SPECIFIC DUTIES. IRON, AND MANUFACTURES OF IRON. Bar iron, rolled or hammered. $15 per ton. Cwt.	Bar iron, rolled or hammered. $15 per ton. Dollars.	Railroad iron, not above six inches high. $12 per ton. Cwt.	Railroad iron, not above six inches high. $12 per ton. Dollars.	Boiler plate. $20 per ton. Cwt.	Boiler plate. $20 per ton. Dollars.	Band iron. $20 per ton. Cwt.	Band iron. $20 per ton. Dollars.	Hoop iron. $20 per ton. Cwt.	Hoop iron. $20 per ton. Dollars.	
1	Sweden and Norway	8,105	30,686									1
2	Swedish West Indies											2
3	Denmark											3
4	Danish West Indies											4
5	Hamburg											5
6	Bremen	5,505	14,069									6
7	Holland											7
8	Dutch West Indies											8
9	Dutch Guiana											9
10	Dutch East Indies											10
11	Belgium											11
12	England	33,174	63,243	48,744	62,226	1,284	2,973	1,915	3,751	5,770	12,942	12
13	Scotland	30	36									13
14	Ireland											14
15	Gibraltar											15
16	Malta											16
17	Canada	4	12			2	8					17
18	Other British North American Possessions	5	12									18
19	British West Indies											19
20	British Honduras											20
21	British Guiana											21
22	British Possessions in Africa											22
23	British East Indies											23
24	France on the Atlantic											24
25	France on the Mediterranean											25
26	French West Indies											26
27	French Guiana											27
28	Spain on the Atlantic											28

Figure 5: Excerpt from the 1859 Foreign Commerce and Navigation Tables. Below each product name, the tariff is listed (e.g. $15 per ton for bar iron, rolled or hammered). Prices are inferred by dividing the 'Dollars' column by the 'Cwt.' column (Cwt. stands for hundredweight, which is equal to 100 pounds).

I deflate all nominal values to 1870 dollars using the Historical Statistics of the United States (2006). Data on county-product output, labor employment, total materials usage and capital are taken from the recently digitized Census of Manufactures for the years 1860 and 1870 (Rotemberg and Hornbeck 2024).

British patent data comes from Hanlon (2016). Table 4 lists the basket of products in my sample alongside British patent counts. As expected, products associated with the two Industrial Revolutions saw more patenting activity, with textiles — the quintessential First

censuses even today, input usage is not recorded by product. See Orr (2022) and De Loecker et al. (2016) for further discussion on inferring input allocations in multi-product establishments. I first compute the firm-specific Leontief share. Then I aggregate from the firm Leontief share to a product Leontief share by taking a weighted average across firms, where the weights are the Domar weight (share of firm revenue in total revenue). Mathematically: Let $\zeta_{z' \to z}$ be the Leontief share of input z' used for z. Let f be an establishment. $\zeta_{z' \to z, f} = \frac{\text{expenditure on input } z' \text{ by } f}{\text{total input expenditure of } f}$, and $\zeta_{z' \to z} = \sum_{f \in z} w_f \zeta_{z' \to z, f}$, where $w_f = \frac{y_f}{\sum_{f' \in z} y_{f'}}$ and y_f is the revenue of establishment f.

Table 4: British Patents by Product Class

Product	British Patents	Category
Cotton Textiles	464	First IR
Worsted Goods	464	First IR
Iron Bar	374	First/Second IR
Iron Pig	374	First/Second IR
Steel	374	Second IR
Paper	301	First IR
Paints	297	Second IR
Glove	222	Neither
Nails	214	First/Second IR
Boots	181	Neither
Hats	169	Neither
Beer	149	Neither
Whiskey	149	Neither
Rope	128	Neither
Glass	95	Neither

Notes: IR = Industrial Revolution. Classification based on whether the product had crucial innovations during the First Industrial Revolution (c. 1760–1840), the Second Industrial Revolution (c. 1870–1914), or neither. Patent data from Hanlon (2016).

Industrial Revolution sector — at the top. Iron and steel, already the second most patented category, foreshadow their dominant role in the coming Second Industrial Revolution. US patent data comes from Shaw (undated, accessed 2025).

$$Dist\_Front_z = \frac{\#\text{ British Patents}_{1855-1859,z}}{\#\text{ US Patents}_{1855-1859,z}}$$

To construct a measure of technological gaps, I harmonize product categories across the two countries' patent data and create a distance to frontier variable, which takes the ratio of British to American patents in the pre-treatment period (1855-1859). This patent-based measure of technological gaps is similar in spirit to the one used by Akcigit, Ates and Impullitti (2024). A value of one means the US has as many patents as the UK (thus it's "at the frontier"), while a large value signifies the US significantly lags behind the UK in that product category. Table 1 displays the rankings of the products by distance to frontier, and Appendix A.2 presents historical context and validation for these rankings as a measure of technological gaps. For example, cotton textiles' near-frontier status reflects decades of transatlantic technology transfer dating to Samuel Slater and Francis Cabot Lowell (Jeremy 1981), while paints' position as the most distant product is consistent with virtually all major innovations in paint originating in Europe during the 19th century (Barnett 2004).

Finally, I extract transportation cost data from Donaldson and Hornbeck (2016) (DH hereafter).[23] This map contains information on whether travel is possible by waterway (fastest), railway (close to fastest) and by wagon (very slow). Assigning values to the speed of travel per mile-ton and assigning a cost to transitioning between different modes of transportation, DH compute the shortest path between any two counties in the US and assign this value as the transportation cost.

I build on DH's network by 1) adding in access to international ports and 2) empirically disciplining the functional form of transportation costs rather than assuming iceberg. I assume that trade operates according to a "hub-and-spoke" system: US port → US internal counties. I first compute the distance from the US county to one of nine US ports[24] using DH's map. Then I set the "transportation cost" to be the minimum of these nine distances, i.e. the transportation cost of any US county and the nearest major port. See Figure 6 for a graphical depiction of the transport network in the East Coast and Midwest. In the next subsection, I empirically discipline the functional form of transportation costs.

My final estimation sample accounts for 27.9% of output, 30.8% of value added, 36.1% of workers, and 32.4% of capital of the total 1860 manufacturing data. The sampled industries are somewhat more capital intensive and produced in larger establishments than manufacturing as a whole, consistent with the fact that products appearing in trade statistics – necessary to construct tariffs and port prices – tend to be tradable and produced at a larger scale. See Appendix Table A13 for further descriptive comparisons.

[23] This was built using Atack's shapefiles (2015, 2016, 2017).

[24] These are ports that had more than $250,000 worth of imports from the United Kingdom in 1870: Boston, New York City, Philadelphia, Baltimore, Charleston, Savannah, New Orleans, Mobile and Galveston.

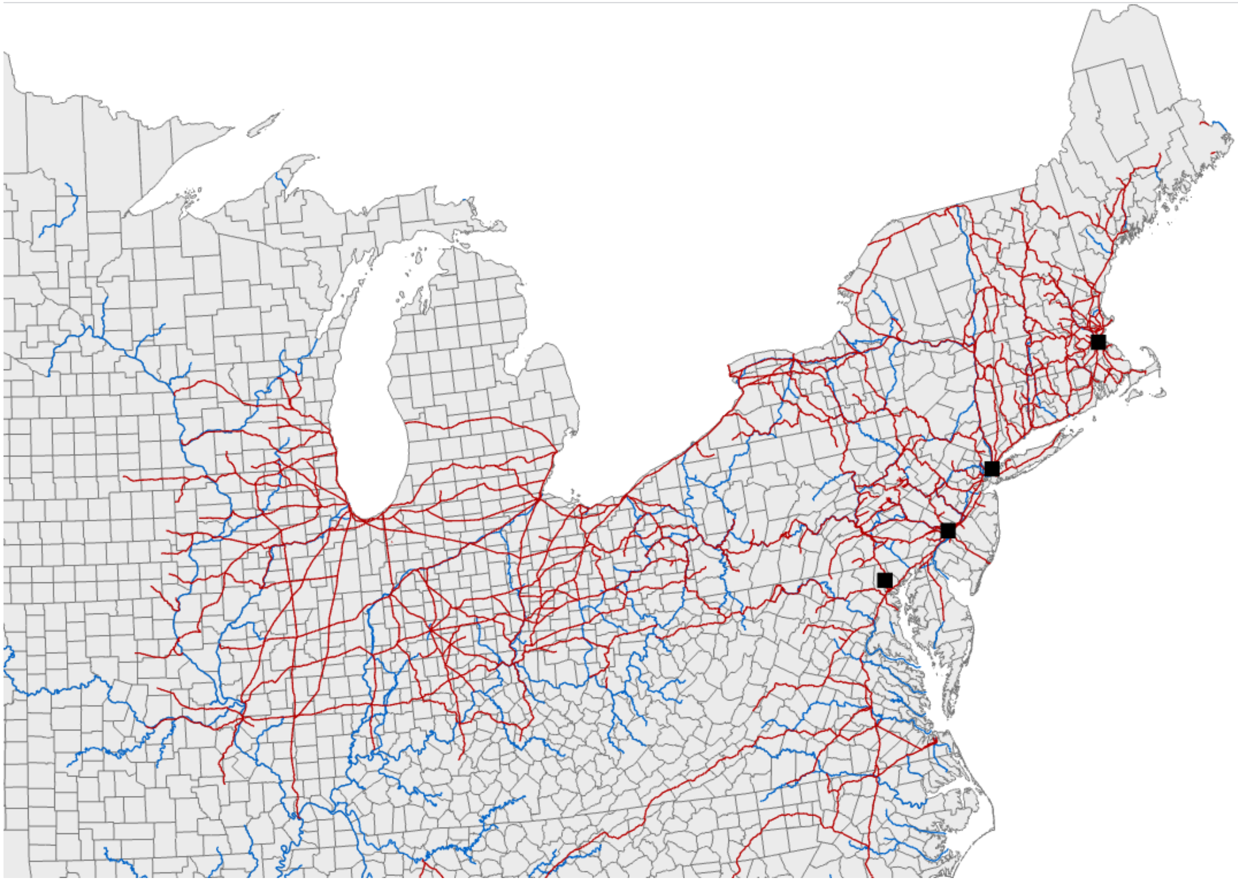

Figure 6: U.S. East Coast and Midwest transportation network, 1860. Railways are red lines, waterways are blue lines. Source: Atack (2015, 2016, 2017). Black squares represent major international ports. See footnote 24 for how major international ports were selected.

5.2 Nature of Transportation Costs

To correctly specify my measure of protection in Section 4.1, I need to measure transportation costs as accurately as possible. I use information in the Aldrich Report (Aldrich 1893) which I digitize specifically for this purpose. Let $\tau_{odzs} = p_{ozs}^{\omega}\zeta_{ods}$, where o is an origin, d is the destination, z is a product, s is year, τ_{odzs} is transportation cost, p_{ozs} is the origin price of product z and ζ_{ods} is the freight rate common to all products.

Note this specification nests the iceberg formulation when $\omega = 1$, which is widely assumed in trade and spatial. In this case, transportation costs scale proportionally with the price of the product in the origin. This may be due to more expensive goods having more expensive handling requirements or due to markup pricing by the transportation sector.[25]

At the other extreme, when $\omega = 0$, transportation costs are fully additive, and are the same for all products regardless of the product's origin price. Thus, only bilateral features between o and d matter, such as physical distance between the two locations and transportation infrastructure. ω can also be between 0 and 1. Taking logs yields:

$$\log \tau_{odzs} = \omega \log p_{ozs} + \delta_{ods} \tag{7}$$

The route-year fixed effect δ_{ods} absorbs everything specific to shipping between o and d in year s (including geographic characteristics and transportation infrastructure), so ω is identified from variation in freight rates across products shipped along the same route in the same year. Critically – and unusually – the Aldrich Report contains information on transportation costs for a given route across different products. Thus τ_{odzs} is directly taken from the data (though only for a subset of routes and products). As for p_{ozs}, I set it to the price of imported products (inclusive of the tariff) if o is a port city.

Table 5: Transportation Cost Regression

	Log Transport Cost
	(1)
Log Import Price ($\hat{\omega}$)	0.139***
	(0.022)
Observations	124
Route-Year FE	Yes

Notes: Regression of log product-specific transportation costs on log import prices with route-year fixed effects, as in equation (7). Transport cost data digitized from the Aldrich Report (1893); port price data from Foreign Commerce and Navigation Tables (1859, 1869, 1879). Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

As Table 5 shows, I find that $\hat{\omega} = 0.139$ with a 95% confidence interval of $[0.0959, 0.182]$, thus transportation costs are much closer to being additive than iceberg. I therefore set $\omega = 0.139$ in the baseline specification, but I examine robustness to $\omega = 0$ (fully additive case), $\omega = 0.0959$ (lower bound of 95% confidence interval), and $\omega = 0.182$ (upper bound of

[25] Hummels and Skiba (2004) show that a monopolist in the transportation sector's markup is increasing in the origin's good price.

95% CI) in Section 6.2.

6 Results and Interpretation

6.1 Output and Patent Results

I now run the specification in equation (3), reproduced here:

$$\Delta \log y_{iz} = \beta[\Delta \log(TP_{iz}) \times Dist\_Front_z] + \theta \Delta \log(TP_{iz}) + \gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz}$$

where i is county, z is product and Δ represents a decade difference. TP_{iz} is the measure of output protection, and X_{iz} is a vector of controls, including the measure of input protection as discussed in Section 4.1. $\Delta\delta_i$ is a county fixed effect and $\Delta\delta_z$ is a product fixed effect. $STS^t_{iz} \Delta \log t_z$ is the IV for $\Delta \log(TP_{iz})$, where $STS^t_{iz} = \frac{t_z p_z}{t_z p_z + \tau_{iz}}$ is the county-product exposure to the nationwide tariff shock, as discussed in Section 4.3.

Table 6: Effect of Trade Protection on Output Growth

	Δ Log Output	
	OLS	IV
	(1)	(2)
Δlog(TP) × Dist_Front	1.359***	1.493***
	(0.411)	(0.398)
Δlog(TP)	−5.144**	−4.424**
	(1.834)	(1.997)
Observations	833	833
Input TP Control	Yes	Yes
County-Year FE	Yes	Yes
Product-Year FE	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	—	15.5

Notes: Each column reports IV estimates of specification 3 of the change in log protection on log output. The instrument is $STS^t_{iz} \times \Delta \log t_z$, the interaction of the tariff share in total protection with the nationwide tariff change. All specifications include controls for the change in input protection, county-year fixed effects and product-year fixed effects. Standard errors clustered at the county and product level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

In Table 6, $\hat{\beta} > 0$ and $\hat{\theta} < 0$. At baseline (i.e. product with distance to frontier of 0 or US technological dominance), a 1% increase in protection leads to a 4.4% fall in output growth. Each unit increase in distance-to-frontier raises this elasticity by 1.5, implying the effect of protection flips from negative to positive at a British-to-US patent ratio of roughly 2.96. To put this in context, protection harms growth for near-frontier products such as cotton textiles (1.31) and iron and steel (2.53), but boosts growth for laggard products such as glass (3.39) and paints (7.82).[26] Appendix C.3 shows that protection exhibits the same frontier-dependent pattern for firm size, value added, input usage and employment.

These results can rationalize the mixed findings in the infant industry literature. Taussig (1915) argues that glass benefited from protection — consistent with my framework, as glass (3.39) exceeds the 2.96 threshold. Irwin (2000a) argues that a large share of the pig iron sector could have survived without protection — again consistent, as pig iron (2.53) falls below the threshold. Irwin and Temin (2001) argue that cotton textiles did not benefit from protection, which my results corroborate given cotton textiles' near-frontier status (1.31).

I also examine whether protection exhibits the same frontier gradient for innovative activity. Using patent-level data from Shaw (undated, accessed 2025), geocoded to the county level using Berkes, Karger and Nencka (2023), I run specification (3) with patent issuances as the dependent variable.

As Column 2 of Table 7 shows, protection exhibits the same frontier-dependent pattern for innovation as for output. At baseline, a 1% increase in protection results in a 7.1% fall in patent issuances. Each unit increase in distance to frontier increases this patent elasticity by 1.3. Protection thus shifts patent activity toward laggard products, suggesting a reshaping of comparative advantage.

[26] The first-stage Kleibergen-Paap F-statistic of 15.5 exceeds the conventional threshold of 10 from Stock and Yogo (2005), though recent work has shown these critical values can be unreliable for valid inference (Montiel Olea and Pflueger, 2013; Lee et al., 2022; Andrews, Stock and Sun, 2019). As a more robust check, I construct weak-instrument-robust confidence sets based on Andrews (2018), implemented using Sun (2018). Both $\hat{\beta}$ and $\hat{\theta}$ remain significant at the 5% level under these tests, which are valid regardless of instrument strength.

Table 7: IV Estimates: Distance to Frontier and Protection Effects on Patent Growth

	Δ Log Patents	
	OLS	IV
	(1)	(2)
Δlog(TP) × Dist_Front	1.175*	1.312**
	(0.596)	(0.572)
Δlog(TP)	-7.459*	-7.138*
	(3.872)	(3.537)
Observations	281	281
Input TP Control	Yes	Yes
County-Year FE	Yes	Yes
Product-Year FE	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)		45.2

Notes: Column (1) reports OLS estimates and column (2) reports IV estimates of specification 3 of the change in log protection on log patents issued. The instrument is $STS_{iz}^{t} \times \Delta \log t_z$. All specifications include controls for the change in input protection, county-year fixed effects and product-year fixed effects. Standard errors clustered at the county and product level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

6.2 Civil War Confounds and Robustness

The Civil War was not merely a fiscal shock — it had wide-ranging impacts across a variety of social and economic domains, making War-related omitted variables loom large. I address this with a battery of robustness checks. My baseline specification excludes the Confederate South due to the potential effects of battles, Confederate governance, Emancipation, and Reconstruction. Results are robust to including the Confederate South in the sample (Column 1 of Table 8). I also examine sensitivity to excluding the four slave states that fought on the side of the Union — Kentucky, Maryland, Missouri, and Delaware; results are robust to their exclusion (Column 2 of Table 8). Finally, I restrict the sample to counties – whether North or South – without a documented Civil War battle from the Civil War Sites Advisory Commission (CWSAC, 1993), thereby removing any county that may have experienced direct physical destruction of productive capacity. Results are once again robust (Column 3 of Table 8).

Beyond direct conflict, the Civil War also generated economic shocks, as military procurement created geographically concentrated demand boosts. I digitize the report of the

Table 8: Robustness: Civil War Sample Restrictions

	Δ Log Output			
	Including South	Excl. Border States	Excl. Battle Counties	Excl. Military Purchases
	(1)	(2)	(3)	(4)
Δlog(TP) $\times$ Dist_Front	1.508***	1.203***	1.568***	1.695***
	(0.406)	(0.267)	(0.406)	(0.549)
Δlog(TP)	-4.159*	-3.634*	-4.691**	-5.073**
	(2.109)	(1.995)	(2.037)	(2.339)
Observations	883	771	853	800
Input TP Control	Yes	Yes	Yes	Yes
County-Year FE	Yes	Yes	Yes	Yes
Product-Year FE	Yes	Yes	Yes	Yes
First-stage F-stat (KP)	16.6	19.2	16.2	13.6

Notes: Each column reports IV estimates of specification 3 of the change in log protection on log output under alternative sample restrictions. Column 1 includes the Confederate South. Column 2 excludes the four Union slave states (Kentucky, Maryland, Missouri, Delaware). Column 3 excludes all counties with a documented Civil War battle (CWSAC, 1993). Column 4 excludes counties with documented military purchases from the Quartermaster General's report. Standard errors clustered at the county and product level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Quartermaster General documenting Civil War purchases by county; Column 4 of Table 8 shows that results are robust to dropping counties with any documented military purchases.

The Civil War also severely disrupted the supply and transportation of cotton, the vast majority of which was produced in the South. If this disruption differentially affected cotton textile producers on the coast relative to interior producers, it could confound the identifying variation. Column 1 of Table 9 shows results are robust to dropping cotton textiles. Appendix Table C15 examines robustness to dropping other products one-by-one.

In addition, I drop the counties containing the five largest cities in 1860 – New York City, Philadelphia, Baltimore, Boston and New Orleans – which may have had particularly large influence over trade policy due to lobbying power or visibility, potentially representing a threat to exogeneity. Columns 2 and 3 of Table 9 respectively show results are robust to dropping these counties, and to additionally dropping their neighboring counties.

Beyond Civil War-related concerns, I also examine robustness to alternative specifications of transportation costs. In Section 5.2, I estimated $\hat{\omega} = 0.139$ and used the following specification for transport costs: $\tau_{iz} = p_z^{0.139}\zeta_i$. In Table 10, results are robust to setting

Table 9: Robustness: Excluding Cotton Textiles and Top 5 Cities

	Δ Log Output		
	Excl. Cotton Textiles	Excl. Top 5 Cities	Excl. Top 5 Cities & Neighbors
	(1)	(2)	(3)
Δlog(TP) $\times$ Dist_Front	1.663***	1.690***	1.527***
	(0.469)	(0.544)	(0.485)
Δlog(TP)	-4.833**	-4.395*	-4.250**
	(2.146)	(2.169)	(1.966)
Observations	743	801	739
Input TP Control	Yes	Yes	Yes
County-Year FE	Yes	Yes	Yes
Product-Year FE	Yes	Yes	Yes
First-stage F-stat (KP)	15.9	15.7	18.1

Notes: Each column reports IV estimates of specification 3 of the change in log protection on log output under alternative sample exclusions. Column 1 excludes cotton textiles. Column 2 excludes the counties containing the five largest cities in 1860 (New York City, Philadelphia, Baltimore, Boston and New Orleans). Column 3 additionally excludes counties neighboring those five. Standard errors clustered at the county and product level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

$\omega = 0$ (corresponding to fully additive transport costs), $\omega = 0.096$ (lower bound of the 95% confidence interval for ω estimated in Section 5.2) and $\omega = 0.182$ (upper bound).

I also examine robustness to specifying distance to frontier in logarithms rather than levels. The log transformation compresses the distribution, reducing the leverage of the most distant products. As Table 11 shows, the interaction remains positive and significant. The implied threshold is $\exp(-\hat{\theta}/\hat{\beta}) = 2.85$, close to the 2.96 implied by the linear specification, indicating that the location of the sign flip is not sensitive to the functional form of the interaction.

6.3 Mechanisms and Alternative Interpretations

The model presented in Section 3 generates the patterns seen in Tables 6 and 7. The key insight is that protection substitutes for innovation as a source of market power for close-to-frontier products. These producers already command a large market share and face sufficiently inelastic residual demand that protection reduces the incremental gain from innovating. For products further behind the frontier, residual demand is more elastic and

Table 10: Robustness to Transportation Cost Specification (ω)

	Δ Log Output		
	$\omega = 0$	$\omega = 0.096$	$\omega = 0.182$
	(1)	(2)	(3)
Δlog(TP) $\times$ Dist_Front	1.066***	1.340***	1.666***
	(0.288)	(0.354)	(0.452)
Δlog(TP)	-3.289*	-4.017**	-4.885**
	(1.557)	(1.820)	(2.225)
Observations	833	833	833
Input TP Control	Yes	Yes	Yes
County-Year FE	Yes	Yes	Yes
Product-Year FE	Yes	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	10.6	14.0	16.9

Notes: Each column reports IV estimates of specification 3 of the change in log protection on log output under alternative values of ω, the elasticity of transportation costs with respect to product price (estimated in Section 5.2). Column 1: fully additive ($\omega = 0$). Column 2: lower bound of 95% confidence interval ($\omega = 0.096$). Column 3: upper bound ($\omega = 0.182$). Standard errors clustered at the county and product level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 11: Effect of Trade Protection on Output Growth: Log Distance to Frontier

	Δ Log Output
	(1)
Δlog(TP) $\times$ log(Dist_Front)	6.156**
	(2.331)
Δlog(TP)	-6.455^{*}
	(3.055)
Observations	833
Input TP Control	Yes
County-Year FE	Yes
Product-Year FE	Yes
First-stage F-statistic (Kleibergen-Paap)	18.0

Notes: The table reports IV estimates of specification 3, replacing $Dist\_Front_z$ with $\log(Dist\_Front_z)$ in the interaction. The instrument is $STS^t_{iz} \times \Delta \log t_z$, the interaction of the tariff share in total protection with the nationwide tariff change. All specifications include controls for the change in input protection, county-year fixed effects and product-year fixed effects. Standard errors clustered at the county and product level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

the negative innovation effect is weaker, allowing the positive demand substitution effect – the shift in demand toward domestic varieties from higher import prices – to dominate. In this section I discuss whether alternative models can generate the same empirical results.

Reallocation According to Static Comparative Advantage

An alternative interpretation comes from the static Ricardian model, which is also consistent with the output results in Table 6. If we interpret $Dist\_Front_z$ as a proxy for US comparative advantage, then protection would reallocate resources from comparative advantage to comparative disadvantage products, generating a similar pattern in output growth.[27] However, the static Ricardian model does not predict differential innovation responses along the technological frontier. As shown in Table 7, protection exhibits the same frontier gradient for patent issuances as for output, suggesting that the pattern of comparative advantage itself is changing rather than merely being reallocated according to existing comparative advantage.

Heterogeneous Trade Elasticities

Next, I rule out an alternative interpretation based on trade elasticities. Standard trade theory predicts that more substitutable products see a larger output response to a given increase in protection. If products far from the frontier also happen to have higher elasticities of substitution, this could rationalize the positive interaction coefficient without any innovation channel as predicted by my model in Section 3.[28] I test this by correlating $Dist\_Front_z$ with trade elasticities from three sources: Caliendo and Parro (2015), Shapiro (2016), and Giri, Yi, and Yilmazkuday (2021).

As Table 12 shows, the correlations are uniformly negative: products far from the frontier have *lower*, not higher, elasticities of substitution, ruling out this alternative interpretation. Results are also robust to directly controlling for $(\sigma_z - 1) \times \Delta \log(TP_{iz})$ in the main specifi-

[27]Trade liberalization, by contrast, reallocates resources from comparative-disadvantage to comparative-advantage products, which is the basis for the gains from trade in a Ricardian setting.

[28]However, this would not be able to rationalize the sign flip or the negative effect of trade protection on output, without adding an additional channel such as innovation or X-inefficiencies.

Table 12: Correlation between Distance to Frontier and Trade Elasticities

Elasticity Source	Correlation with $Dist\_Front_z$
Caliendo & Parro (2015)	-0.229
Shapiro (2016)	-0.555
Giri, Yi & Yilmazkuday (2021)	-0.170
Median	-0.533

Notes: Each row reports the correlation between the distance-to-frontier measure ($Dist\_Front_z$) and the trade elasticty ($\sigma_z - 1$) from the indicated source. All correlations are negative, indicating that products further from the British technological frontier are less substitutable — the opposite of what would be required for trade elasticities to explain the positive interaction coefficient in Table 6.

cation (Appendix Table C14).

Pricing-to-Market

The empirical framework defines total protection as tariffs plus transportation costs expressed in ad valorem form, which is theoretically consistent with the model in Section 3. However, the British exporter charges a destination-varying markup $\mu_{iz} = \frac{1+(\sigma-1)\lambda_{iz}}{(\sigma-1)\lambda_{iz}}$, which is decreasing in the domestic share λ_{iz} and hence in the level of protection in county i, representing incomplete passthrough.[29] Strictly speaking, this markup enters the error term as an omitted variable in the main specification, Equation (3). Can such pricing-to-market behavior on its own account for the empirical results without an innovation channel?

Two arguments make this unlikely. First, pricing-to-market on its own cannot deliver a negative effect of protection on output. For output to fall, the exporter's markup would have to increase when tariffs rise, representing *negative* passthrough, which is very rarely observed empirically and difficult to motivate theoretically. My framework predicts positive and incomplete passthrough, i.e. the British firm cuts its markup when tariffs rise to dampen the loss in market share. The majority of other models also predict positive passthrough, whether complete or incomplete. More fundamentally, with innovation shut off, the decomposition in Equation (1) leaves only the substitution effect, which is strictly positive: a static

[29] Equivalently, we can think of this as an American importer buying from the British exporter at the port, paying the tariff to the Federal government, then shipping the product to the interior while charging a destination-specific markup. The analysis is the same.

markup response dampens the output response but cannot reverse its sign.

Second, and more decisively, a framework with no innovation channel makes no predictions for patenting. As Table 7 shows, protection shifts patent activity with the same frontier gradient as output. Thus, mechanically, pricing-to-market alone without an innovation channel cannot generate this pattern.

Quality Upgrading via Alchian-Allen Effects

Another interpretation arises from the Alchian-Allen effect (Alchian and Allen 1964, Hummels and Skiba 2004), where additive transport costs lower the relative cost of high-quality varieties.[30] In the baseline Alchian-Allen logic, high-τ_{iz} locations should receive relatively higher-quality imports. An ad valorem tariff applies proportionally to the product price, eroding the shipping advantage that additive transport costs give high-quality goods. Higher tariffs therefore push the foreign import mix toward lower-quality varieties, an effect strongest in high-τ_{iz} interior markets (Hummels and Skiba 2004).[31]

However, while the Alchian-Allen effect clearly predicts changes in the import quality mix after tariffs are applied, it does not make a prediction for the output response of interior firms. First, while some interior firms may upgrade quality to escape competition, others contract as competition at the low end intensifies, with the total effect on county output ambiguous. Second, even for firms that do move upmarket, this does not guarantee a higher output response. A firm that upgrades its quality sells higher priced units, but it may reduce total quantities produced, resulting in an ambiguous effect on output. In addition, there is no clear reason, regardless of its sign, why this effect should be different for far-from-frontier items relative to close-to-frontier items.

[30] To see this in my notation, take low- and high-quality varieties of product z, with prices $p_z^L < p_z^H$. Total protection is $TP_{iz}(p) = t_z + \tau_{iz}(p)/p$. Since I estimate transportation costs as $\tau_{iz}(p) = p^{\omega}\zeta_i$, with $\omega = 0.139 < 1$, the transport component is $\tau_{iz}(p)/p = \zeta_i p^{\omega-1}$. Hence $\tau_{iz}(p_z^H)/p_z^H < \tau_{iz}(p_z^L)/p_z^L$, so $TP_{iz}(p_z^H) < TP_{iz}(p_z^L)$: high-quality varieties face lower proportional total protection than low-quality varieties.

[31] Let $a_H \equiv \tau_{iz}(p_z^H)/p_z^H$ and $a_L \equiv \tau_{iz}(p_z^L)/p_z^L$, with $a_H < a_L$. The relative delivered price of the high-quality variety is $(p_z^H/p_z^L)\{[(1+t_z)+a_H]/[(1+t_z)+a_L]\}$. Since $\partial\{[(1+t_z)+a_H]/[(1+t_z)+a_L]\}/\partial t_z = (a_L - a_H)/[(1+t_z)+a_L]^2 > 0$, an increase in the ad valorem tariff raises the relative delivered price of high-quality foreign varieties.

Factor Migration and Spatial Spillovers

The baseline specification treats county-product pairs as independent, but protection in one pair may affect outcomes in another. Such spillovers violate the stable unit treatment value assumption (SUTVA) and may bias $\hat{\beta}$ and $\hat{\theta}$. Can these omitted spillovers account for the empirical results? First, we discuss factor migration. Suppose that under constant returns to scale, more protected county-product pairs draw factors, workers for example, away from less protected ones. Treated pairs then grow partly at the expense of control pairs, widening the estimated gap and biasing $\hat{\theta}$ upward.

Second, we discuss demand spillovers. When a county becomes more protected, local consumers may substitute not only toward producers in their own county, but also toward producers in other counties. Let this omitted spillover term equal $S_{iz} = \sum_{j \neq i} \omega_{ijz} \Delta \log TP_{jz}$, where ω_{ijz} measures the bilateral spatial link between county i and j for product z. This term enters with a positive coefficient, since protection elsewhere raises demand for i's output. The sign of this bias is governed by $\mathrm{Cov}(\Delta \log TP_{iz}, S_{iz})$. This covariance is likely positive: coastal counties i have stronger links to other coastal counties j, and those counties receive the largest protection shock, raising S_{iz}. Demand spillovers therefore bias $\hat{\theta}$ upward as well.

Under either channel the negative estimates for close-to-frontier products are conservative: the true θ is more negative still. A standard gravity model cannot rationalize a negative response of output to protection, without adding in additional channels such as variable markups and innovation as in this paper.

6.4 Structural Interpretation of Regression Coefficients

Having ruled out leading alternative models, I now use the model in Section 3 to structurally interpret the estimated coefficients in Section 6.1. In Appendix D.2, I show the finite change in output with respect to a finite change in total protection can be written, to a first-order, as:

$$\frac{\Delta \log Y_{iz}}{\Delta \log(1+TP_{iz})} \approx \underbrace{(\sigma-1)(1-\lambda_{iz})}_{\text{Gravity benchmark}} \cdot \underbrace{\frac{\sigma+(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}{\sigma^2-(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}}_{\text{Bertrand dampening}\in(0,1)} \cdot \left(1+\log\alpha \cdot \frac{\Delta I_{iz}}{\Delta \log(1+TP_{iz})}\right) \tag{8}$$

The first term, $(\sigma-1)(1-\lambda_{iz})$, is the expenditure-switching elasticity implied by standard gravity models: output responds more to protection when import penetration $(1-\lambda_{iz})$ is high, as there is more expenditure to redirect toward the local variety,[32] and when products are more substitutable (higher σ). See Appendix B.5 for a derivation of this expression in a gravity setting. As low τ_{iz} locations (i.e. close to port locations) will tend to have larger baseline import penetration $1-\lambda_{iz}$ for a given product, they will see larger output increases via this channel.

The second term is a Bertrand dampening factor that is strictly between zero and one. This factor captures strategic markup adjustment: when protection raises the foreign firm's delivered cost, part of this increase is absorbed by endogenous markup adjustments rather than wholly translating into expenditure switching. In standard gravity frameworks, markups are constant, so this dampening term is equal to 1.

Finally, the term inside the parentheses governs whether the net effect is positive or negative: the total effect turns negative when $\frac{\Delta I_{iz}}{\Delta \log(1+TP_{iz})} < -\frac{1}{\log\alpha}$, that is, when protection reduces innovation sufficiently to overwhelm the mechanical substitution effect. Proposition 2 and Corollary 2 in Section 3 characterize when this occurs as a function of distance to the technological frontier.

As Appendix C.1 of Borusyak and Hull (2021) shows, in the presence of heterogeneous treatment effects, an IV regression identifies a convex weighted average of unit-level causal effects under a suitable monotonicity condition, which holds mechanically in my setting due to the definitions of $\Delta \log TP_{iz}$ and $STS^t_{iz}\Delta \log t_z$. Suppose we run the following regression:

[32] To see this in the extreme case, suppose a location has no foreign imports, i.e. $1-\lambda_{iz}=0$. In this case, an increase in protection is redundant under this framework, as there are no imports to substitute away from towards the local variety.

$$\Delta \log y_{iz} = \sum_z \beta_z[\Delta \log(1 + TP_{iz})] + \gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz}$$

with $STS^t_{iz}\Delta \log t_z$ as an IV for $\Delta \log(1 + TP_{iz})$. According to Borusyak and Hull Appendix C.1 (2021), this $\hat{\beta}_z$ estimates a variance-weighted average of the county-level treatment effects:

$$\hat{\beta}_z \xrightarrow{p} \frac{\sum_i \beta_{iz}\, w^2_{iz}}{\sum_i w^2_{iz}} \tag{9}$$

where $w^2_{iz} = Var(STS^t_{iz}\Delta \log t_z \mid \Delta\delta_i, \Delta\delta_z)$ is the variance of the instrument conditional on fixed effects. Under the model in Section 3, $\beta_{iz} \approx \frac{\Delta \log Y_{iz}}{\Delta \log(1+TP_{iz})}$, which is unpacked in Equation (8) above in this Section. The w^2_{iz} weights are larger for coastal counties, which have high STS^t_{iz} and therefore more instrument variation. Therefore, $\hat{\beta}_z$ recovers a convex weighted average of β_{iz}s, with the responses of coastal areas weighted more heavily (i.e. areas more affected by tariff increases).

The main empirical specification in Equation (3) then linearly projects β_z onto distance to the frontier: $\beta_z = \theta + \beta \cdot Dist\_Front_z$. Therefore, $\hat{\beta}$ measures how a convex, coastal-weighted average of county-level treatment effects varies linearly with distance to the technological frontier.

7 Welfare and General Equilibrium

In Section 3, I presented a model in which the effect of trade protection on output and innovation depends on distance to the technological frontier. This was a partial equilibrium result: aggregate objects such as income and wages were held fixed. This partial equilibrium model maps naturally to the empirical specification in Equation (3), where county fixed effects $\Delta\delta_i$ absorb changes in local income and wages; the estimated coefficients should therefore be interpreted as partial equilibrium effects.

However, trade protection can also affect aggregate income and wages, as emphasized by the "missing intercept" or "micro-to-macro" literature (Chodorow-Reich (2019, 2020) and Guren et al. (2021)). For example, real incomes can fall due to higher import prices or rise due to increased demand from protection, affecting the output of all products. Thus, for a full accounting of the effects of trade protection, I incorporate general-equilibrium channels in this section and use the resulting framework to study the welfare implications of trade. I derive an expression that nests the famous Arkolakis, Costinot and Rodriguez-Clare (ACR 2012) result and allows for welfare losses from trade.

Section 7.1 presents the model. Section 7.2 presents market clearing conditions and the definition of equilibrium. Finally, Section 7.3 studies the welfare effects of trade under this model.

7.1 Model Setup

I build on the product-market environment from Section 3. The key difference is that I now allow aggregates such as expenditures and wages to change. In this setting, granular oligopolistic firms internalize their impact on aggregates, posing technical difficulties as noted by Neary (2016). I therefore assume a nested demand structure, as in Atkeson and Burstein (2008). For each product z, let submarkets be indexed by $\omega_z \in [0, 1]$.[33] In each submarket ω_z, a US firm and a UK firm compete à la Bertrand in the US market, and the US firm chooses whether to innovate before setting prices, as in Section 3. Thus, while firms are large and granular within their own submarket, each submarket is measure zero in the aggregate economy. Therefore, each individual firm cannot affect aggregates, though collectively their behavior shapes aggregate incomes and wages.

We have three nests in this model. I assume upper-tier preferences across products z are

[33] For example, within product category z = cotton textiles, one submarket is white shirts while another is black dresses.

Cobb-Douglas with expenditure share χ_z, so

$$E_z = \chi_z E, \qquad \sum_z \chi_z = 1.$$

where E is total expenditure. Within product z, the product-level quantity aggregator is Cobb–Douglas across submarkets $\omega_z \in [0,1]$:

$$\log Q_z = \int_0^1 \log Q_z(\omega_z) d\omega_z.$$

where $Q_z(\omega_z)$ is consumption in submarket ω_z and Q_z is aggregate consumption of product z. The corresponding product-level price index is

$$\log P_z = \int_0^1 \log P_z(\omega_z) d\omega_z.$$

Within each submarket ω_z, the US and UK varieties are CES substitutes:

$$Q_z(\omega_z) = \left[q_z^{US}(\omega_z)^{\frac{\sigma-1}{\sigma}} + q_z^{UK}(\omega_z)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 1.$$

The submarket price index is

$$P_z(\omega_z) = \left[p_z^{US}(\omega_z)^{1-\sigma} + p_z^{UK}(\omega_z)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}.$$

7.1.1 Submarket Firm Problem.

Let E_{ω_z} denote expenditure in submarket ω_z, which satisfies $E_z = \int_0^1 E_{\omega_z} d\omega_z$. In each submarket ω_z, the US firm chooses whether to innovate,

$$I_z(\omega_z) \in \{0, 1\}.$$

The marginal cost of the US firm under no innovation and innovation is, respectively:

$$c_z^{US}(0) = \frac{w^{US}}{T_z^{US}}, \qquad c_z^{US}(1) = \frac{w^{US}}{\alpha T_z^{US}}, \qquad \alpha > 1.$$

The UK firm has delivered marginal cost

$$c_z^{UK} = (1 + TP_z)\frac{w^{UK}}{T_z^{UK}}.$$

Total protection TP_z is determined by tariffs and transportation costs, as defined in Section 4.1. Given innovation status $I_z(\omega_z)$, firms compete in prices within the submarket. Let

$$\lambda_z(\omega_z) = \left(\frac{p_z^{US}(\omega_z)}{P_z(\omega_z)}\right)^{1-\sigma}$$

denote the US expenditure share in submarket ω_z. The residual demand elasticity faced by the US firm is:

$$\varepsilon_z^{US}(\omega_z) = \sigma - (\sigma - 1)\lambda_z(\omega_z).$$

Thus the US firm's optimal price satisfies

$$p_z^{US}(\omega_z) = \frac{\varepsilon_z^{US}(\omega_z)}{\varepsilon_z^{US}(\omega_z) - 1} c_z^{US}(I_z(\omega_z)). \tag{10}$$

Similarly, the UK firm's residual demand elasticity is

$$\varepsilon_z^{UK}(\omega_z) = \sigma - (\sigma - 1)(1 - \lambda_z(\omega_z)),$$

and its optimal price satisfies

$$p_z^{UK}(\omega_z) = \frac{\varepsilon_z^{UK}(\omega_z)}{\varepsilon_z^{UK}(\omega_z) - 1} c_z^{UK}. \tag{11}$$

Submarket sales and operating profits of the US firm are, respectively,

$$X_z^{US}(\omega_z) = E_{\omega_z}\lambda_z(\omega_z), \qquad \Pi_z^{op}(\omega_z) = \frac{E_{\omega_z}\lambda_z(\omega_z)}{\varepsilon_z^{US}(\omega_z)}.$$

The submarket US firm innovates if and only if the gain from innovation exceeds the fixed cost:

$$I_z(\omega_z) = \mathbf{1}\{\Pi_z^{op}(\omega_z \mid I_z(\omega_z) = 1) - \Pi_z^{op}(\omega_z \mid I_z(\omega_z) = 0) \geq F\}. \tag{12}$$

7.1.2 Aggregation Across Submarkets

Within each product z, all submarkets ω_z share the same primitives: T_z^{US}, T_z^{UK}, t_z, w^{US}, w^{UK}, F, and α. I restrict attention to pure symmetric equilibria, in which all submarkets within product z choose the same innovation status:

$$I_z(\omega_z) = I_z \qquad \forall \omega_z \in [0,1]. \tag{13}$$

Let $G_z^s \equiv G_z(I_{z,-\omega_z} = s)$ denote the gain from innovation when all other submarkets choose $s \in \{0,1\}$.[34] I impose a strategic substitutability condition in innovation incentives:

$$G_z^0 > G_z^1 \qquad \forall z.$$

Namely, the gain from innovating is strictly higher when other submarkets do not innovate than when they do. This assumption guarantees uniqueness of the pure symmetric equilibrium.

The unique innovation choice I_z, along with common primitives, implies $p_z^{US}(\omega_z) = p_z^{US}$, $p_z^{UK}(\omega_z) = p_z^{UK}$, $P_z(\omega_z) = P_z$ and $\lambda_z(\omega_z) = \lambda_z$ for all ω_z. In addition, sales aggregate:

[34] Existence of a pure symmetric equilibrium is guaranteed if for all z:

$$G_z^0 \leq F \qquad \text{or} \qquad G_z^1 \geq F.$$

$$Y_z = \int_0^1 \lambda_z(\omega_z) E_{\omega_z} d\omega_z = \int_0^1 \lambda_z E_{\omega_z} d\omega_z = \lambda_z \int_0^1 E_{\omega_z} d\omega_z = \lambda_z E_z.$$

7.2 General Equilibrium

This economy is closed by three clearing conditions. First, goods market clearing for each product z requires output to equal expenditure times market share:

$$Y_z = E_z \lambda_z. \tag{14}$$

Second, aggregate income equals labor income and profits:

$$Y = w^{US} L + \Pi. \tag{15}$$

where w^{US} is the wage, $L = \sum_z L_z$ is the labor endowment, and $\Pi = \sum_z \Pi_z$ are net profits earned across all products, where $\Pi_z = \Pi_z^{op} - FI_z$.

Third, aggregate expenditure satisfies:

$$E = Y + TR + B. \tag{16}$$

where TR is tariff revenue and B is an exogenous deficit financed by borrowing.

Given fundamentals, technology, tariffs, and transportation costs $\{\{T_z^{US}, T_z^{UK}, \chi_z, t_z, \tau_z\}_z, F, B, L, w^{UK}\}$, an equilibrium is a vector of prices, innovation choices, and wage $\{\{p_z^{US}, p_z^{UK}, I_z\}_z, w^{US}\}$ such that firms optimize according to the pricing conditions in Equations (10) and (11), innovation choices satisfy Equation (12), and the clearing conditions in Equations (14), (15), and (16) hold.

7.3 Welfare

We now study the welfare implications of trade through the lens of this model. I define welfare as the ratio of total expenditure to the price index:

$$W = \frac{E}{P}$$

where $P = \prod_z P_z^{\chi_z}$ and $E = \sum_z E_z$. I begin with the one-product case. We assume that there is a small shock to transportation costs $d \log \tau$, which has no revenue effects. In Appendix D.4.4 I study a shock to tariffs $d \log t$, which includes an additional term accounting for revenue effects. Note λ is the share of expenditure spent on the domestic variety. Around an equilibrium, the first-order welfare change is:

$$d \log W = \underbrace{\frac{1}{1-\sigma} d \log \lambda}_{\text{ACR term}} + \underbrace{dI \left(\log \alpha - \frac{F}{L + B - FI} \right)}_{\text{innovation effect}}.$$

See Appendix D.4.2 for the derivation of this expression. The first term is the classic ACR domestic-share term (Arkolakis, Costinot, and Rodriguez-Clare, 2012), which fully summarizes welfare changes in standard gravity frameworks. A larger fall in the domestic share leads to larger gains in welfare, with the exact mechanism depending on the particular microfoundation.[35]

My framework has an additional term. Unlike in standard gravity frameworks, the change in domestic expenditure share is no longer a sufficient ex-post statistic for the welfare gains from trade. It also matters whether the trade shock changes the firm's innovation decision, beyond its effect on the domestic expenditure share.

This additional term also allows for the possibility of welfare losses from moving toward trade. Trade can reduce the incentive to innovate, i.e. $dI < 0$. If in addition, the step size is larger than the innovation resource cost (i.e. $\log \alpha > \frac{F}{L+B-FI}$), signifying that innovation

[35] In Krugman (1980), for instance, the gains arise from love of variety, while in a Ricardian setting (Eaton and Kortum 2002) they arise from specialization according to comparative advantage.

has net positive social value, the innovation effect is negative and may dominate the ACR gains from trade term, resulting in a net welfare loss from trade. Appendix D.4.1 provides a numerical example of a welfare loss from moving toward trade. However, a sufficient (but not necessary) condition for welfare gains from trade is if ex-post innovation is unaffected, that is $dI = 0$. In such a scenario, the formula reduces to the ACR case, and a move towards trade always increases welfare.

How does distance to frontier affect the welfare gains from trade? A high distance to frontier has two effects pulling in opposite directions: On the one hand, when the foreign producer is far ahead, the domestic share will decline more dramatically after a move towards free trade, increasing the ACR gains from trade term. On the other hand, as Proposition 1 in Section 3 suggests, the possibility of a decrease in innovation is higher for far from frontier products relative to close to frontier products when trade is liberalized.

We then turn to an ex-ante perspective. Relative to the gravity models covered in Arkolakis et al. (2012), this model tends to predict lower welfare gains for high distance-to-frontier products after a shock to trade costs (assuming innovation has net positive social value). Two channels push in this direction. First, endogenous markup adjustments dampen the change in the domestic share relative to a constant-markup model, attenuating the ACR term (see Equation (8)). Second, a move toward trade can trigger a cut in innovation, subtracting further from the gains and potentially reversing their sign. However, for close-to-frontier products the comparison is ambiguous: the first channel still attenuates the ACR term, but liberalization raises the return to innovation for these products, so the innovation effect works in the opposite direction.

Next, we turn to a multiproduct version of the gains from trade formula, presented below:

$$
\begin{aligned}
d\log W = & \underbrace{\sum_z \chi_z \frac{1}{1-\sigma} d\log\lambda_z}_{\text{ACR term}} + \underbrace{\sum_z dI_z \left[\chi_z \log\alpha - \frac{F}{E}\right]}_{\text{innovation effect}} \\
& - \underbrace{\sum_z \chi_z d\log\mu_z}_{\text{procompetitive effect}} + \underbrace{\sum_z \chi_z \frac{\Pi_z^{op}}{E_z} d\log\Pi_z^{op}}_{\text{rent effect}}
\end{aligned}
$$

where $\Pi_z^{op} = \Pi_z + FI_z$ is operating profit, which I defined to be equal to net profit plus the fixed cost (if paid), and $\mu_z = \frac{\varepsilon_z^{US}}{\varepsilon_z^{US}-1}$ is the markup. See Appendix D.4.3 for a proof of this expression. The first two terms are the multiproduct analogues of the corresponding single-product terms, weighted by Cobb-Douglas expenditure shares χ_z. Two additional terms arise, capturing reallocation across products. The procompetitive effect captures changes in markups: if trade lowers markups, i.e. $d\log\mu_z < 0$, it raises welfare by lowering consumer prices. This effect is larger when the product is a larger share of consumption (high χ_z). The rent effect captures changes in domestic profits, which under my assumptions are rebated to the representative consumer and therefore raise welfare when $d\log\Pi_z^{op} > 0$. This effect is larger when the product is a larger share of expenditure, χ_z, and when it has a larger profit share, Π_z^{op}/E_z. In the one-product case, these two effects exactly cancel: an increase in markups raises consumer prices but is offset exactly by the increase in profits rebated to the consumer. In the multiproduct case, this cancellation no longer holds product by product because markup changes are weighted by expenditure shares, while profit changes are weighted by profit intensity times expenditure shares. Thus markup increases are welfare reducing when they occur in products that are important for consumption, but welfare enhancing when they occur in products that are important sources of domestic profits. Because of these additional terms, we do not have a simple sufficient condition for welfare gains from trade as in the single product case. The formula with tariff-revenue effects can be found in Appendix D.4.4.

How does distance to frontier affect these two terms? Both terms are larger in magnitude (but with opposite signs) for close-to-frontier products than far-from-frontier products: close-to-frontier firms see larger cuts in markups but also larger reductions in rents, as they command larger market shares and face more inelastic residual demand, giving them both more market power to lose and more rents to lose. Therefore, if close-to-frontier products tend to have high profit intensity relative to their expenditure shares, the rent losses from liberalization dominate the procompetitive gains, reducing the net welfare benefit of trade for those products. Whether the procompetitive or rent effect dominates in the aggregate therefore depends on the joint distribution of expenditure shares and profit intensities across the frontier spectrum.

8 Conclusion

This paper establishes both empirically and theoretically that the effect of protection is negative for products near the technological frontier, yet positive for products further behind, using new data and a novel empirical design addressing various threats to exogeneity. It also establishes that protection can alter innovation incentives in both the negative and the positive direction, and the net welfare implications of protection are ambiguous.

What do these results mean for the practice of trade and industrial policy today? First, an important caveat: as discussed in Section 2 and Appendix A.1, US trading partners did not retaliate against American tariffs in this period. Today, retaliatory tariffs are the norm rather than the exception, meaning that protection comes at the additional cost of reduced export market access — a cost absent from my estimates. I therefore interpret the results as an upper bound for the efficacy of trade protection in the modern era.

Second, the model suggests, if protectionist tariffs are to be used, that they be used in a carefully targeted manner. This framework suggests Liberation Day tariffs imposed by the second Trump administration – which applied uniform country-wide tariff rates based on bilateral trade deficits – are too blunt an instrument. For example, the 32% tariff applied to Taiwan made no distinction between advanced semiconductors, iron bolts and tapioca starch used to produce boba tea.

On the other hand, the Biden administration imposed very high 100% tariffs on Chinese electric vehicles (EVs) in its last year in office. Whether these tariffs are successful according to the framework in this paper is dependent on the relative quality and productivity of Chinese EVs relative to US EVs. Chinese battery cell prices are over 20% lower than American ones and EV production costs are over 30% lower (IEA 2025), and some analysts argue that Chinese EV manufacturers have equaled or surpassed their American counterparts in product quality and technological capability (Ezell 2024). If these assessments are correct, my framework suggests such targeted tariffs could raise the return to innovation and boost growth for American EV producers.

While this paper focuses on technological gaps, other determinants of competitiveness also condition the innovation and output response to protection, such as wages and transport costs. For example, a developing country with sufficiently low wages or poor international port infrastructure may be insulated from foreign competition despite a large technological gap, so that liberalization triggers a domestic innovation gain. Conversely, if wages are not low enough to compensate for a large technological gap, liberalization poses greater risks to domestic output and innovation.

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Appendix A Additional Historical Context

Appendix A.1 Why Didn't Major Trade Partners Retaliate?

The three major European powers – the United Kingdom, France, and Prussia (which was to become Germany) – did not retaliate against the US for their high tariffs.

The United Kingdom, the industrial leader in this period, had committed to free trade in the mid 19th century, and begun eliminating tariffs on all goods except for sin goods such as alcohol. This had occurred after a bitter political battle over The Corn Laws[36], a set of protectionist measures designed to protect domestic British agrarian producers, which lead to food riots and arguably contributed to the Irish famine. In 1846, the British Prime Minister, Robert Peel, was able to overcome opposition and repealed the Corn Laws. This set into motion a process that ended up reducing tariffs on all goods other than sin goods by 1860. Furthermore, retaliatory measures to high American tariffs, though debated by the British especially during the Civil War period, did not end up passing.

In the mid-19th century, France begun a process of bilateral trade reductions with partner countries. There is no documented active bilateral treaty between France and the US in 1860-1870 (Pahre 2008). Furthermore, there is no evidence that the US was a major consideration of the general trade policy for France (i.e. tariffs applied to countries not subject to a specific bilateral treaty). Generally, it appears France, unlike the UK, was mostly concerned with trade with other European nations and its colonial empire, not the US.

Finally, the German states had yet to unify (which had occurred in 1871), inhibiting unified responses to tariff policy. For example, the two major German ports, Hamburg and Bremen, were not part of the Zollverein customs union, and thus not subject to the tariff policy of the rest of Germany. In addition, every member in the Zollverein had a veto till 1867. Finally, just like France, it appeared that the US was not a major consideration of the German states. Rather, it seems the focus of German trade policy was with other European states.

Appendix A.2 Validation of Distance to Frontier Measure

This appendix provides historical evidence that the patent-based distance to frontier measure captures meaningful technological differences across products.

Cotton textiles (1.31). Despite restrictions on technology transfers, British emigrants such as Samuel Slater — referred to as the "father of American manufactures" (White 1836) — alongside Americans like Francis Cabot Lowell, who memorized British power loom designs during a visit to England, established the New England textile sector. By the early 19th century, technology flowed in both directions across the Atlantic (Jeremy 1981), consistent with near-frontier status.

Iron and steel (2.53). Allen (1979): "In the middle of the nineteenth century Britain was the major supplier of iron and steel to the world market, while Germany and the United States were substantial importers... I shall demonstrate that Britain's mid-century export success was due to its superior technical efficiency." Temin (1964) adds that before the Civil

[36]The Economist magazine was founded as an anti-Corn Law journal. David Ricardo, in his capacity as a member of parliament, had also advocated against the Corn Laws.

War, American manufacturers produced only small quantities of steel, as they were "unable to master the demanding requirements to create steel through puddling," with imports from Sheffield dominating the domestic market.

Beer (2.87). Pattinson (2014): "While at the beginning of the century, British breweries had led the world in technology and scale of operation, by the end of it the Americans were the ones at the bleeding edge of innovation" — consistent with mid-range distance to frontier in the 1850s.

Glass (3.39). Scoville (1944): "The first glassmen in the United States naturally patterned their methods after those current in European countries, and they continued to do so throughout most of the nineteenth century. Few innovations were of American origin. Technological development lagged appreciably behind that in Belgium, England, France, and Germany."

Paints (7.82). Virtually all major pigment innovations of the nineteenth century originated in Europe (Barnett 2004). Leading British paint firms were founded by chemists—Berger Paints by a Frankfurt-trained chemist in 1760, Winsor & Newton by a chemist-artist partnership in 1832. By contrast, the first American paint firm to employ a chemist was Sherwin Williams in 1884.

Appendix A.3 Sample Characteristics

Table A13: Characteristics of the Matched Sample Relative to Manufacturing, 1860

Measure	All Manufacturing	Sample
Capital share	0.405	0.448
Labor share	0.147	0.160
Materials share	0.447	0.392
Output per establishment	14,427.48	21,560.35
Workers per establishment	9.66	18.52
Output per worker	1,493.37	1,164.28
Value added per establishment	6,256.82	10,319.28
Value added per worker	647.64	557.25
Capital per worker	766.22	694.16

Table A13 compares the characteristics of my sample to those of the broader manufacturing sector. The two are broadly similar, though the industries in my sample are somewhat more capital intensive and are produced in larger establishments on average. This is consistent with the fact that products appearing in trade statistics, which are necessary to construct tariffs and port prices, are more likely to be tradable and produced at larger scale than more locally oriented manufacturing activities.

Appendix B Treatment Construction and Identification Results

Appendix B.1 First Order Approximation of $\Delta \log(TP_{iz})$

$$\Delta \log(TP_{iz}) \approx d\log(TP_{iz}) = \frac{1}{TP_{iz}} d(TP_{iz}) = \frac{1}{TP_{iz}} d\left(t_z + \frac{\tau_{iz}}{p_z}\right)$$

Focusing on the last term:

$$d\left(\frac{\tau_{iz}}{p_z}\right) = \frac{1}{p_z^2}\left(p_z\, d\tau_{iz} - \tau_{iz}\, dp_z\right) = \frac{1}{p_z^2}\left(p_z\, \tau_{iz}\, d\log \tau_{iz} - \tau_{iz}\, p_z\, d\log p_z\right) =$$

$$\frac{\tau_{iz}}{p_z}\left(d\log \tau_{iz} - d\log p_z\right)$$

Plugging this back:

$$d\log(TP_{iz}) = \frac{1}{TP_{iz}}\left(d(t_z) + \frac{\tau_{iz}}{p_z}\left(d\log \tau_{iz} - d\log p_z\right)\right)$$

Note:

$$\frac{1}{TP_{iz}} d(t_z) = \frac{t_z}{t_z + \frac{\tau_{iz}}{p_z}} d\log t_z = \frac{p_z\, t_z}{\underbrace{p_z\, t_z + \tau_{iz}}_{STS_{iz}^t}} d\log t_z$$

Also note:

$$\frac{1}{TP_{iz}}\left(\frac{\tau_{iz}}{p_z}\left(d\log \tau_{iz} - d\log p_z\right)\right) = \frac{\tau_{iz}}{\underbrace{t_z\, p_z + \tau_{iz}}_{STS_{iz}^\tau}}\left(d\log \tau_{iz} - d\log p_z\right)$$

Putting these two together:

$$d\log(TP_{iz}) = STS_{iz}^t\, d\log t_z + STS_{iz}^\tau\left(d\log \tau_{iz} - d\log p_z\right)$$

Appendix B.2 Proof of Sufficiency of Identification Condition

The IV orthogonality condition is:

$$\mathbb{E}(STS_{iz}^t \Delta \log t_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z) = 0$$

I show the following orthogonality condition implies the above:

$$\mathbb{E}(\Delta \log t_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z, STS_{iz}^t) = 0$$

Proof (Borusyak and Hull 2023):

By Law of Iterated Expectations,

$$\mathbb{E}(STS^t_{iz}\Delta\log t_z \cdot \Delta e_{iz}|\Delta\delta_i,\Delta\delta_z) = \mathbb{E}(\mathbb{E}(STS^t_{iz}\cdot\Delta\log t_z\cdot\Delta e_{iz}|STS^t_{iz},\Delta\delta_i,\Delta\delta_z))$$

$$= \mathbb{E}(STS^t_{iz}\cdot\mathbb{E}(\Delta\log t_z\cdot\Delta e_{iz}|STS^t_{iz},\Delta\delta_i,\Delta\delta_z))$$

If $\mathbb{E}(\Delta\log t_z\cdot\Delta e_{iz}|STS^t_{iz},\Delta\delta_i,\Delta\delta_z) = 0$, we yield the desired result

Appendix B.3 Elasticity of Revenue to the Tariff

Revenue $R = ptM$, where p is price of imports, t is tariff and M is quantity of imports.

$$\frac{\partial R}{\partial 1+t} = \underbrace{\frac{\partial t}{\partial 1+t}}_{1} pM + \frac{\partial M}{\partial 1+t} tp + tM\frac{\partial p}{\partial 1+t}$$

Elasticity of revenue to tariff is equal to:

$$\epsilon^R_t := \frac{\partial R}{\partial 1+t}\frac{1+t}{ptM} = \frac{pM(1+t)}{ptM} + \frac{\partial M}{\partial 1+t}\frac{tp(1+t)}{ptM} + \frac{\partial p}{\partial 1+t}\frac{tM(1+t)}{ptM}$$

Focusing on the middle term:

$$\frac{\partial M}{\partial 1+t}\frac{tp(1+t)}{ptM} = \frac{\partial M}{\partial 1+t}\frac{1+t}{M} = \frac{\partial\log M}{\partial\log 1+t} = \frac{\partial\log M}{\partial\log p(1+t)}\frac{\partial\log p(1+t)}{\partial\log(1+t)} = \epsilon^D_t\rho$$

where ϵ^M_d is the import elasticity of demand and ρ is the passthrough of tariffs to the (tariff-inclusive) price.

Focusing on the last term:

$$\frac{\partial p}{\partial 1+t}\frac{tM(1+t)}{ptM} = \frac{\partial p}{\partial 1+t}\frac{(1+t)}{p} = \frac{\partial\log p(1+t)}{\partial\log 1+t} - 1$$

where in the second equality we used the result that $\frac{\partial p}{\partial 1+t} = (\frac{\partial p(1+t)}{\partial 1+t} - p)\frac{1}{1+t}$ (Below is derivation of this)

$$\frac{\partial p(1+t)}{\partial(1+t)} = p + (1+t)\frac{\partial p}{\partial(1+t)} \iff \frac{\partial p}{\partial(1+t)} = \frac{1}{1+t}(\frac{\partial p(1+t)}{\partial(1+t)} - p)$$

Putting everything together, we yield:

$$\epsilon^R_t = \frac{1+t}{t} + \rho(\epsilon^D_t + 1) - 1$$

Appendix B.4 Formulas Including Specific Tariffs

In the main text, for brevity and ease of exposition I present tariffs as if they are all ad valorem. In the empirics, I take into account whether tariffs are ad valorem or specific.

Here, I present the formulas taking into acccount specific tariffs.

The pricing equation with specific tariffs now is:

$$p_{iz} = p_z + t_z^{AV} p_z + t_z^{SP} + \tau_{iz}$$

where t_z^{AV} denotes ad-valorem tariff, and t_z^{SP} denotes specific tariffs.

The ad valorem equivalent of tariffs and tranpsortation cost is:

$$TP_{iz} := \frac{p_{iz} - p_z}{p_z} = t_z^{AV} + \frac{t_z^{SP}}{p_z} + \frac{\tau_{iz}}{p_z}$$

Finally, the first-order approximation in this case will be:

$$d\log(TP_{iz}) = STS_{iz}^{t^{AV}} d\log t_z^{AV} + STS_{iz}^{\tau}(d\log \tau_{iz} - d\log p_z) + STS_{iz}^{t^{SP}}(d\log t_z^{SP} - d\log p_z)$$

where $STS_{iz}^{t^{SP}} = \frac{t_z^{SP}}{t_z^{AV} p_z + t_z^{SP} + \tau_{iz}}$

Appendix B.5 Spatial Impact of Tariffs

This appendix seeks to show that the impact of a tariff is larger in coastal areas than interior areas via the lens of a gravity equation.

Appendix B.5.1 Gravity Equation

Consider the following gravity equation. The gravity equation can be microfounded by Anderson (1979), Krugman (1980), Eaton and Kortum (2002), and Melitz (2003):[37]

$$X_{iiz} = p_{iiz}^{1-\sigma} P_{iz}^{\sigma-1} E_{iz}$$

where i indexes county, z indexes product. X_{iiz} is sales of county i product z in county i, p_{iiz} is the local price, P_{iz} is the local price index, E_{iz} is total expenditure in county i, and $\sigma > 1$ is the elasticity of substitution.

Let preferences across products be Cobb-Douglas. Thus, $E_{iz} = \chi_{iz} E_i$, where E_i is total expenditure in county i and χ_{iz} is the Cobb-Douglas preference weight county i places on z goods. Note this allows different counties to have different preferences across products.

The local price index aggregates domestic and foreign (imported) varieties:

$$P_{iz} = (p_{iiz}^{1-\sigma} + p_{ifz}^{1-\sigma})^{\frac{1}{1-\sigma}}$$

where p_{ifz} is the delivered price of foreign goods to county i. Log-differentiating the gravity equation:

[37] See Head and Mayer (2014) for discussion on these microfoundations and others.

$$d\log X_{iiz} = (1-\sigma)d\log p_{iiz} + (\sigma-1)d\log P_{iz} + d\log E_{iz}$$

As gravity frameworks use CES demand and assume either perfect or monopolistic competition as the market structure, markups are constant, and domestic prices do not adjust to changes in foreign prices. Thus, a tariff shock, which directly affects the price of foreign products, does not affect domestic prices in the gravity framework: $d\log p_{iiz} = 0$. The change in price index is equal to:

$$d\log P_{iz} = \frac{p_{iiz}^{1-\sigma}}{P_{iz}^{1-\sigma}} d\log p_{iiz} + \frac{p_{ifz}^{1-\sigma}}{P_{iz}^{1-\sigma}} d\log p_{ifz}$$

Let λ_{ifz} equal county i's expenditure share on foreign imports:

$$\lambda_{ifz} = \frac{X_{ifz}}{E_{iz}} = \frac{p_{ifz}^{1-\sigma} P_{iz}^{\sigma-1} E_{iz}}{E_{iz}} = p_{ifz}^{1-\sigma} P_{iz}^{\sigma-1} = \frac{p_{ifz}^{1-\sigma}}{P_{iz}^{1-\sigma}}$$

Thus:

$$d\log P_{iz} = \lambda_{ifz} \cdot d\log p_{ifz}$$

Intuitively, the price index in county i rises more in counties with larger import shares λ_{ifz} and for areas where the foreign price shock $d\log p_{ifz}$ is larger. Both of these tend to be higher for low-τ_{iz} or coastal areas. Finally, I assume preferences do not change in this period, so $d\log \chi_{iz} = 0$. This leaves changes in total expenditures in county i, $d\log E_i$, which will be absorbed by the county fixed effect $\Delta\delta_i$, signifying the estimate captures partial equilibrium responses.

$$d\log E_{iz} = \underbrace{d\log \chi_{iz}}_{0} + \underbrace{d\log E_i}_{\Delta\delta_i}$$

Putting everything together, we have:

$$d\log X_{iiz} = (\sigma-1)\lambda_{ifz} \cdot d\log p_{ifz}$$

The delivered price of foreign goods incorporates both the factory-gate price, tariffs, and transport costs:

$$p_{ifz} = p_{fz}(1+t_z) + \tau_{iz} = p_{fz} \cdot (1 + TP_{iz}), \qquad TP_{iz} \equiv t_z + \frac{\tau_{iz}}{p_{fz}}$$

where TP_{iz} is total protection, the ad valorem equivalent of the combined tariff and transport wedge. Taking the differential with $d\log p_{fz} = 0$:

$$d\log p_{ifz} = d\log(1 + TP_{iz})$$

Finally for simplicity we assume the county only sells to itself (no exports). Thus, $Y_{iz} = X_{iiz}$. Therefore, the change in output can be written as:

$$d\log Y_{iz} = (\sigma-1)\lambda_{ifz} \cdot d\log(1 + TP_{iz})$$

When nationwide tariffs go up ($d \log t_z > 0$), the change in protection, i.e. $d \log TP_{iz}$, is larger for low-τ_{iz} than high-τ_{iz} locations. In addition, the foreign import share λ_{ifz} tends to be higher for low-τ_{iz} areas relative to high-τ_{iz} areas, further amplifying the effect. Thus, the predicted change in output in a gravity framework will be larger for coastal i than for interior i. According to appendix C.1 of Borusyak and Hull (2021), under a monotonicity condition, the base specification estimates a convex weighted average, where weights are proportional to λ_{ifz} (locations with higher λ_{ifz}, higher import shares, are weighed more heavily in these calculations).

Appendix B.6 Comparison with Shift-Share Design

This appendix formally contrasts the shift-share design and the design in my paper. For ease of exposition, I omit controls other than fixed effects from this discussion.

A shift-share design is interested in an outcome at the regional level, e.g. county level. For example, a researcher may be interested in how output Y_i in county i is affected by its import exposure. A standard shift-share in this literature (i.e. the local labor market approach) is defined as:

$$Z_i^{SS} = \sum_z s_{iz} \Delta g_z,$$

where s_{iz} is the baseline share of product z employment in county i, and Δg_z is an aggregate shock to product z. For example, Δg_z could be changes in nationwide product tariffs t_z, as in this paper. Or Δg_z could be growth in Chinese imports in product z as in Autor, Dorn and Hanson (2013). The shift-share regression is:

$$\Delta Y_i = \beta Z_i^{SS} + \Delta e_i.$$

By contrast, the specification in my paper is:

$$\Delta Y_{iz} = \beta \Delta \log(TP_{iz}) + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz},$$

where Y_{iz} is manufacturing output in county i and product z, $\Delta \log(TP_{iz})$ is the change in trade protection in county i and product z, and $\Delta\delta_i$ and $\Delta\delta_z$ are county and product fixed effects.

I instrument $\Delta \log(TP_{iz})$ with

$$Z_{iz} = STS_{iz}^t \Delta g_z,$$

where STS_{iz}^t is the share of the tariff in total protection as defined in Section 4.3. In this paper, $\Delta g_z = \Delta \log t_z$ where t_z is the nationwide tariff on product z.

Shift-share designs compare regions with different industrial compositions after a nationwide product-level shock, whereas my design compares the same product across counties with differential increases in protection due to differences in transportation costs to international ports. Put differently, the shift-share regression studies outcome changes at the regional level i, whereas my regression studies outcome changes at the region-product level iz. This allows me to include product fixed effects and compare across counties within the same product.

These two approaches rely on different identification assumptions. As shown by Equation 6, the identification condition for my design is:

$$\mathbb{E}\left(\Delta g_z \cdot \Delta e_{iz} \mid STS^t_{iz}, \Delta\delta_i, \Delta\delta_z\right) = 0.$$

Exogenous Shocks. Shift-share identification arguments typically rely on one of two justifications. I first consider the exogenous-shocks view of Borusyak, Hull and Jaravel (2022), which requires:

$$\mathbb{E}\left(\Delta g_z \cdot \Delta \bar{e}_z \cdot s_z\right) = 0,$$

where $s_z = \sum_i s_{iz}$, and

$$\Delta \bar{e}_z = \frac{\sum_i s_{iz}\Delta e_i}{\sum_i s_{iz}}.$$

Shift-share designs require the national shock Δg_z to be uncorrelated with **national** product-level shocks $\Delta \bar{e}_z$; my design instead requires the national shock Δg_z to be uncorrelated with **local** county-product shocks Δe_{iz}. In this sense my identification condition is more granular.[38] In addition, my design allows for the inclusion of product fixed effects $\Delta\delta_z$, unlike shift-share designs. Therefore, any product-wide shock common to all regional units i, such as technological breakthroughs or federal policy changes, may confound shift-share designs, while being absorbed by product fixed effects in my design.

Exogenous Shares. Now I compare my design to the exogenous-shares view of Goldsmith-Pinkham, Sorkin and Swift (2020):

$$\mathbb{E}\left(s_{iz}\Delta e_i\right) = 0.$$

The identification condition here requires that initial industrial composition is uncorrelated with future location shocks. In this sense, within the shift-share family, the exogenous-shares view is more granular than the exogenous-shocks view: identification comes from the interaction of a region-product variable, s_{iz}, with a regional shock, Δe_i, rather than from the interaction of a product-level shock, Δg_z, with a product-level residual, $\Delta \bar{e}_z$. However, this shifts the identifying assumption onto the determinants of baseline industrial composition. In the trade context, this assumption is demanding: regions specializing in different products may also be differentially exposed to non-trade shocks, such as immigration flows, technological change, unionization drives, or region-specific industrial decline.

[38] In some contexts, where the nationwide shock Δg_z is a policy lever such as tariffs, this condition is arguably more plausible because violating it requires a more specific form of targeting. For example, policymakers targeting fast-growing products nationally would threaten shift-share designs, but would not threaten my design unless they targeted products based on differential trends across counties within the same product, which is a higher informational and cognitive burden on policymakers.

Appendix C Additional Empirical Results

Appendix C.1 Robustness to Trade Elasticity

Table C14: Robustness: Controlling for Trade Elasticity

	Δ Log Output			
	(1)	(2)	(3)	(4)
	Median	CP (2015)	Shapiro (2016)	GYY (2021)
Δlog(AVE) × Distance to Frontier	2.650**	1.585***	3.586**	1.995***
	(1.027)	(0.433)	(1.370)	(0.695)
$(\sigma_z - 1) \times \Delta\log(\text{AVE})$	0.851	1.426	0.993*	0.395
	(0.602)	(2.030)	(0.539)	(0.491)
Δlog(AVE)	−15.009*	−9.242	−23.539*	−9.253
	(7.914)	(6.883)	(11.410)	(6.052)
Observations	833	833	833	833
Input AVE Control	Yes	Yes	Yes	Yes
County-Year FE	Yes	Yes	Yes	Yes
Product-Year FE	Yes	Yes	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	7.59	3.55	25.62	7.71

Notes: Each column reports IV estimates of specification 3 of the change in log protection on log output, augmented with an interaction between the trade elasticity of substitution $(\sigma_z - 1)$ and the change in log protection. Each column uses a different source for σ_z: Column 1 uses the median across sources, Column 2 uses Caliendo and Parro (2015), Column 3 uses Shapiro (2016), Column 4 uses Giri, Yi, and Yilmazkuday (2021). The distance-to-frontier interaction remains significant across all specifications. Standard errors clustered at the county and product level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Appendix C.2 Dropping Products

Here we run our baseline specification, but we drop products one-by-one.

Table C15: Leave-One-Out Robustness Check

Product Excluded	$\Delta \log(\text{TP}) \times$ *Dist_to_Front*	$\Delta \log(\text{TP})$	KP F	N
Beer	1.823***	−5.839**	12.09	614
	(0.468)	(2.354)		
Boots	1.640***	−4.589*	16.00	592
	(0.451)	(2.315)		
Cotton Goods	1.659***	−4.821**	15.82	743
	(0.468)	(2.142)		
Glass	1.502***	−4.342*	21.42	797
	(0.428)	(2.060)		
Glove	1.443***	−4.359*	15.63	826
	(0.408)	(2.035)		
Hats	1.350***	−3.721*	16.40	794
	(0.387)	(2.052)		
Iron Bar	1.348**	−4.017*	13.51	766
	(0.471)	(2.092)		
Iron Pig	1.174**	−2.967**	12.20	774
	(0.516)	(1.321)		
Nails	1.394***	−4.511**	10.17	808
	(0.395)	(1.870)		
Paints	1.938**	−5.751*	13.83	823
	(0.881)	(2.833)		
Paper	1.528***	−4.287**	15.55	716
	(0.421)	(2.120)		
Rope	1.113***	−3.066*	15.44	798
	(0.289)	(1.525)		
Steel	1.503***	−4.465**	15.44	830
	(0.403)	(2.012)		
Whiskey	1.674***	−5.935**	15.65	729
	(0.503)	(2.355)		
Worsted Goods	1.498***	−4.430**	15.45	830
	(0.400)	(2.004)		

Notes: Standard errors in parentheses. Each row reports IV coefficient estimates from a regression excluding the indicated product. TP denotes total protection (tariff plus transportation cost). KP F is the Kleibergen-Paap rk Wald F-statistic. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Appendix C.3 Additional Outcome Variables

Table C16: Trade Protection and Firm Size, Value Added and Factor Usage

	(1) Δ Log Output per Estab.	(2) Δ Log Value Added	(3) Δ Log Labor	(4) Δ Log Materials	(5) Δ Log Capital
Δlog(AVE) $\times$ Distance to Frontier	1.818***	1.015**	1.383**	1.905***	−0.675
	(0.482)	(0.413)	(0.530)	(0.475)	(0.498)
Δlog(AVE)	−6.049***	−3.588	−6.896**	−6.076**	−0.969
	(1.976)	(2.061)	(2.560)	(2.235)	(2.712)
Observations	833	832	833	833	833
Input AVE Control	Yes	Yes	Yes	Yes	Yes
County-Year FE	Yes	Yes	Yes	Yes	Yes
Product-Year FE	Yes	Yes	Yes	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	15.5	15.6	15.5	15.5	15.5

Notes: Standard errors two-way clustered by county and product in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Appendix C.4 Extensive Margins

I run the following specification:

$$\mathbf{1}[y_{iz,1860} = 0,\ y_{iz,1870} > 0] = \beta[\Delta \log(TP_{iz}) \times Dist\_Front_z] + \theta\Delta \log(TP_{iz}) +$$
$$\gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz}$$

The dependent variable is an indicator function taking the value of 1 if a county-product pair did not produce in 1860 but has produced in 1870. The right hand side variables are the same as main text.

We find no statistically significant differential effect of trade protection on entry by distance to the frontier. This suggests that my model, which purely operates on the intensive margin, is the appropriate framework for this setting.

Table C17: Extensive Margin

	$1[y_{iz,1860} = 0,\ y_{iz,1870} > 0]$	
	OLS	IV
	(1)	(2)
Δlog(TP) $\times$ Dist_Front	-0.002	-0.007
	(0.013)	(0.022)
Δlog(TP)	0.033	0.082
	(0.067)	(0.126)
Observations	11,501	11,501
Input TP Control	Yes	Yes
County-Year FE	Yes	Yes
Product-Year FE	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	—	20.31

Appendix D Additional Theoretical Results

Appendix D.1 Numerical Example

Take

$$\sigma = 8, \qquad E_z = 1, \qquad D_z = 1.05, \qquad \alpha = 3, \qquad TP_z^0 = 0.25, \qquad TP_z^1 = 0.35, \qquad F = 0.4343$$

Solving the Bertrand-CES share equation gives

$$G_z(D_z, TP_z^0) = 0.4371,$$

and

$$G_z(D_z, TP_z^1) = 0.4316.$$

Then

$$G_z(D_z, TP_z^0) > F > G_z(D_z, TP_z^1),$$

so the firm innovates before protection but not after:

$$I_z^0 = 1, \qquad I_z^1 = 0.$$

Before the protection increase, the firm innovates and captures a 92.4% market share:

$$Y_z^0 = E_z \lambda_z \left(\frac{3 \times 1.25}{1.05} \right) = \lambda_z(3.57) = 0.9242.$$

After the protection increase, the firm no longer innovates and its market share falls to 66.1% despite higher protection:

$$Y_z^1 = E_z \lambda_z \left(\frac{1.35}{1.05} \right) = \lambda_z(1.29) = 0.6610.$$

Thus $Y_z^1 < Y_z^0$: output falls by 28.5%. Protection eliminates the incentive to innovate, and the resulting loss of competitiveness more than offsets the direct protective effect.

Appendix D.2 Structural Derivation of Elasticity Term

We start from the finite decomposition in Equation (1). To obtain a crisp form for the structural elasticity, we take a first-order approximation around the pre-protection equilibrium. This gives

$$\frac{\Delta \log Y_{iz}}{\Delta \log(1+TP_{iz})} \approx \left.\frac{\partial \log Y_{iz}}{\partial \log(1+TP_{iz})}\right|_0 + \left.\frac{\partial \log Y_{iz}}{\partial \log T_z^{US}}\right|_0 \cdot \frac{\Delta \log T_z^{US}}{\Delta I_{iz}^*} \cdot \frac{\Delta I_{iz}^*}{\Delta \log(1+TP_{iz})}.$$

The derivation proceeds in three steps: (i) showing that the substitution and technology elasticities are equal, (ii) computing the technology step $\Delta \log T_z^{US}/\Delta I_{iz}^*$, and (iii) deriving the common elasticity under Bertrand-CES competition.

Step 1: Equality of Elasticities

US output is $Y_{iz} = E_z \lambda_{iz}(r_{iz})$, where relative competitiveness is

$$r_{iz} = \frac{\alpha^{I_{iz}}(1+TP_{iz})}{D_z}.$$

and $D_z = T_z^{UK}/T_z^{US}$. Equivalently,

$$\log r_{iz} = I_{iz}\log\alpha + \log(1+TP_{iz}) + \log T_z^{US} - \log T_z^{UK}.$$

Since $\log Y_{iz} = \log E_z + \log \lambda_{iz}(r_{iz})$, by the chain rule:

$$\frac{\partial \log Y_{iz}}{\partial \log(1+TP_{iz})} = \frac{d\log\lambda_{iz}}{d\log r_{iz}} \cdot \underbrace{\frac{\partial \log r_{iz}}{\partial \log(1+TP_{iz})}}_{=1} = \frac{d\log\lambda_{iz}}{d\log r_{iz}}.$$

$$\frac{\partial \log Y_{iz}}{\partial \log T_z^{US}} = \frac{d\log\lambda_{iz}}{d\log r_{iz}} \cdot \underbrace{\frac{\partial \log r_{iz}}{\partial \log T_z^{US}}}_{=1} = \frac{d\log\lambda_{iz}}{d\log r_{iz}}.$$

Step 2: Technology Step

When the US firm innovates, its technology improves from T_z^{US} to αT_z^{US}. Therefore:

$$\frac{\Delta \log T_z^{US}}{\Delta I_{iz}^*} = \log(\alpha T_z^{US}) - \log(T_z^{US}) = \log\alpha.$$

Step 3: Derivation of $d\log\lambda_{iz}/d\log r_{iz}$

Under Bertrand-CES competition, the US and UK prices are:

$$p_z^{US} = \mu^{US}(\lambda_{iz}) \cdot c_z^{US}, \qquad p_z^{UK} = \mu^{UK}(\lambda_{iz}) \cdot c_z^{UK}.$$

where

$$\mu^{US}(\lambda_{iz}) = \frac{\varepsilon_z^{US}}{\varepsilon_z^{US}-1} = \frac{\sigma-(\sigma-1)\lambda_{iz}}{(\sigma-1)(1-\lambda_{iz})}$$

and

$$\mu^{UK}(\lambda_{iz}) = \frac{\varepsilon_z^{UK}}{\varepsilon_z^{UK}-1} = \frac{1+(\sigma-1)\lambda_{iz}}{(\sigma-1)\lambda_{iz}}.$$

The CES expenditure share satisfies:

$$\log\frac{\lambda_{iz}}{1-\lambda_{iz}} = (\sigma-1)\left[\log r_{iz} + \log\mu^{UK}(\lambda_{iz}) - \log\mu^{US}(\lambda_{iz})\right].$$

Implicitly differentiating with respect to $\log r_{iz}$:

$$\frac{1}{\lambda_{iz}(1-\lambda_{iz})}\frac{d\lambda_{iz}}{d\log r_{iz}} = (\sigma-1)\left[1+\left(\frac{d\log\mu^{UK}}{d\lambda_{iz}} - \frac{d\log\mu^{US}}{d\lambda_{iz}}\right)\frac{d\lambda_{iz}}{d\log r_{iz}}\right].$$

Computing the markup elasticities:

$$\frac{d\log\mu^{US}}{d\lambda_{iz}} = \frac{1}{\varepsilon_z^{US}(1-\lambda_{iz})}, \qquad \frac{d\log\mu^{UK}}{d\lambda_{iz}} = \frac{-1}{\lambda_{iz}\varepsilon_z^{UK}}.$$

Substituting and solving for $d\lambda_{iz}/d\log r_{iz}$:

$$\frac{d\log\lambda_{iz}}{d\log r_{iz}} = \frac{(\sigma-1)(1-\lambda_{iz})}{1+\frac{(\sigma-1)(1-\lambda_{iz})}{\varepsilon_z^{UK}}+\frac{(\sigma-1)\lambda_{iz}}{\varepsilon_z^{US}}}.$$

Substituting $\varepsilon_z^{US} = \sigma-(\sigma-1)\lambda_{iz}$ and $\varepsilon_z^{UK} = 1+(\sigma-1)\lambda_{iz}$, and letting $a = (\sigma-1)\lambda_{iz}$ and $b = (\sigma-1)(1-\lambda_{iz})$ so that $a+b = \sigma-1$, the denominator becomes:

$$1+\frac{b}{1+a}+\frac{a}{1+b} = \frac{(1+a)(1+b)+b(1+b)+a(1+a)}{(1+a)(1+b)} = \frac{1+2(a+b)+a^2+b^2+ab}{(1+a)(1+b)}.$$

Using $a^2+b^2 = (a+b)^2-2ab$:

$$= \frac{(1+a+b)^2-ab}{(1+a)(1+b)} = \frac{\sigma^2-(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}{(1+a)(1+b)}.$$

And

$$(1+a)(1+b) = 1+a+b+ab = \sigma+(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz}).$$

Therefore:

$$\frac{d\log\lambda_{iz}}{d\log r_{iz}} = (\sigma-1)(1-\lambda_{iz})\cdot\frac{\sigma+(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}{\sigma^2-(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}.$$

Combining

Substituting Steps 1–3 into the first-order approximation above gives:

$$\begin{aligned}\frac{\Delta\log Y_{iz}}{\Delta\log(1+TP_{iz})} &\approx \frac{d\log\lambda_{iz}}{d\log r_{iz}} + \frac{d\log\lambda_{iz}}{d\log r_{iz}}\cdot\log\alpha\cdot\frac{\Delta I^*_{iz}}{\Delta\log(1+TP_{iz})}\\ &= \frac{d\log\lambda_{iz}}{d\log r_{iz}}\cdot\left(1+\log\alpha\cdot\frac{\Delta I^*_{iz}}{\Delta\log(1+TP_{iz})}\right).\end{aligned}$$

Using the expression for $d\log\lambda_{iz}/d\log r_{iz}$ derived above:

$$\frac{\Delta\log Y_{iz}}{\Delta\log(1+TP_{iz})} \approx (\sigma-1)(1-\lambda_{iz})\cdot\frac{\sigma+(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}{\sigma^2-(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}\cdot\left(1+\log\alpha\cdot\frac{\Delta I^*_{iz}}{\Delta\log(1+TP_{iz})}\right),$$

which yields Equation (8). ■

Appendix D.3 Comparative Statics Results

Appendix D.3.1 Proof of Proposition 1

Proof. We have

$$G_{iz}(D_z, TP^1_{iz}) = G_{iz}\left(\frac{1+TP^0_{iz}}{1+TP^1_{iz}}D_z,\ TP^0_{iz}\right).$$

Above-threshold result. Since $r\to 0$ as $D_z\to\infty$, there exists $\bar{D}$ such that both $(1+TP^0_{iz})/D_z$ and $(1+TP^1_{iz})/D_z$ lie in $(0,\bar{r}/\alpha)$ for all $D_z\geq\bar{D}$, where $\bar{r}$ is given by Lemma A1 below. Writing $u=\log r$ and $\psi(u)=\Pi^{op}(e^u)$, the gain from innovation is

$$G_{iz} = \psi(u+\log\alpha)-\psi(u).$$

By Lemma A1, ψ is strictly convex on this region, so $\psi'(u+\log\alpha)>\psi'(u)$ and hence

$$\frac{\partial G_{iz}}{\partial u} = \psi'(u+\log\alpha)-\psi'(u) > 0.$$

Thus G_{iz} is strictly increasing in r, and since r is increasing in TP_{iz},

$$G_{iz}(D_z,TP^1_{iz}) \geq G_{iz}(D_z,TP^0_{iz}) \qquad \text{for all } D_z\geq\bar{D}.$$

Because $I^*_{iz}=\mathbf{1}\{G_{iz}\geq F_z\}$ is nondecreasing in G_{iz}, it follows that $I^*_{iz}(D_z,TP^1_{iz})\geq I^*_{iz}(D_z,TP^0_{iz})$, and therefore

$$\Delta I^*_{iz}\geq 0.$$

Let $D^{I,*}_{iz}$ denote the smallest $\bar{D}$ for which this holds on $[\bar{D},\infty)$.

Below-threshold result. Since Π^{op} is strictly increasing and $\alpha > 1$, $G_{iz} > 0$ for all D_z. As $D_z \to 0$, relative competitiveness diverges, $\lambda \to 1$ and $\Pi^{op} \to E_z$ at both r and αr, so $G_{iz} \to 0$: the US firm captures the market with or without innovating. As $D_z \to \infty$, $\lambda \to 0$ and $\Pi^{op} \to 0$ at both arguments, so again $G_{iz} \to 0$: the firm is uncompetitive either way. A continuous, strictly positive function vanishing at both ends of $(0, \infty)$ attains an interior maximum. Hence there is a region of r on which G_{iz} is strictly decreasing, and therefore strictly decreasing in TP_{iz}:

$$G_{iz}(D_z, TP^1_{iz}) < G_{iz}(D_z, TP^0_{iz}).$$

For any such D_z, any fixed cost satisfying

$$G_{iz}(D_z, TP^1_{iz}) < F_z \leq G_{iz}(D_z, TP^0_{iz})$$

yields $I^*_{iz}(D_z, TP^0_{iz}) = 1$ and $I^*_{iz}(D_z, TP^1_{iz}) = 0$, so

$$\Delta I^*_{iz} = -1.$$

Appendix D.1 exhibits parameter values for which this holds. ■

Lemma A1. There exists $\bar{r} > 0$ such that Π^{op} is strictly convex in $\log r$ on $(0, \bar{r})$.
Proof. Write $\psi(u) \equiv \Pi^{op}(e^u)$ and let $\varepsilon(u)$ denote the elasticity of operating profit with respect to competitiveness,

$$\varepsilon(u) \equiv \left.\frac{d \log \Pi^{op}}{d \log r}\right|_{r=e^u} = \underbrace{\frac{\sigma}{\sigma - (\sigma-1)\lambda}}_{d \log \Pi^{op}/d \log \lambda} \cdot \underbrace{(\sigma-1)(1-\lambda)\,\frac{\sigma + (\sigma-1)^2\lambda(1-\lambda)}{\sigma^2 - (\sigma-1)^2\lambda(1-\lambda)}}_{d \log \lambda / d \log r},$$

where the second factor is derived in Appendix D.2. Then $\psi' = \varepsilon\psi$ and

$$\psi'' = \left(\varepsilon' + \varepsilon^2\right)\psi.$$

As $u \to -\infty$ we have $\lambda \to 0$, so

$$\varepsilon \;\longrightarrow\; \frac{\sigma}{\sigma} \cdot (\sigma - 1) \cdot \frac{\sigma}{\sigma^2} = \frac{\sigma - 1}{\sigma} \equiv \nu > 0.$$

Moreover ε is a smooth function of λ with bounded derivative near $\lambda = 0$, while $d\lambda/du = \lambda \cdot (d \log \lambda / d \log r) = O(\lambda) \to 0$; hence $\varepsilon' \to 0$. Therefore

$$\frac{\psi''}{\psi} \;\longrightarrow\; \nu^2 > 0,$$

and since $\psi > 0$, there exists $\bar{r} > 0$ such that $\psi'' > 0$ on $(0, \bar{r})$. ■

Appendix D.3.2 Proof of Proposition 2

Proof. *Above-threshold result.* Let $I^0_{iz} = I^*_{iz}(D_z, TP^0_{iz})$ and $I^1_{iz} = I^*_{iz}(D_z, TP^1_{iz})$. For $D_z \geq D^{I,*}_{iz}$, Proposition 1 gives $I^1_{iz} \geq I^0_{iz}$. Since $TP^1_{iz} > TP^0_{iz}$ and $\alpha > 1$,

$$\frac{\alpha^{I^1_{iz}}(1 + TP^1_{iz})}{D_z} > \frac{\alpha^{I^0_{iz}}(1 + TP^0_{iz})}{D_z}.$$

Because the Bertrand-CES expenditure share $\lambda(\cdot)$ is strictly increasing in relative US competitiveness,

$$Y^1_{iz} = E_z\lambda\left(\frac{\alpha^{I^1_{iz}}(1 + TP^1_{iz})}{D_z}\right) > E_z\lambda\left(\frac{\alpha^{I^0_{iz}}(1 + TP^0_{iz})}{D_z}\right) = Y^0_{iz},$$

so $\Delta \log Y_{iz} > 0$. Since this holds for all $D_z \geq D^{I,*}_{iz}$, the smallest threshold above which output rises satisfies $D^{Y,*}_{iz} \leq D^{I,*}_{iz}$.

Below-threshold result. Suppose the parameters are such that $\Delta I^*_{iz} = -1$, which Proposition 1 shows is possible, and suppose in addition that

$$\alpha > \frac{1 + TP^1_{iz}}{1 + TP^0_{iz}},$$

so that $\alpha(1 + TP^0_{iz}) > 1 + TP^1_{iz}$: the loss of the innovation step more than offsets the direct increase in protection. The firm innovates at TP^0_{iz} but not at TP^1_{iz}, so, since $\lambda(\cdot)$ is strictly increasing,

$$Y^0_{iz} = E_z\lambda\left(\frac{\alpha(1 + TP^0_{iz})}{D_z}\right) > E_z\lambda\left(\frac{1 + TP^1_{iz}}{D_z}\right) = Y^1_{iz},$$

and therefore $\Delta \log Y_{iz} < 0$. ■

Appendix D.3.3 Proof of Corollary 1

Proof. By single-peakedness, the set of products that innovate at the initial protection level is an interval:

$$\{D_z : G_{iz}(D_z, TP^0_{iz}) \geq F_z\} = [D_L, D_H].$$

Since D_z and TP_{iz} enter only through $(1 + TP_{iz})/D_z$, the corresponding set at the higher protection level is obtained by scaling by $(1 + TP^1_{iz})/(1 + TP^0_{iz})$:

$$\{D_z : G_{iz}(D_z, TP^1_{iz}) \geq F_z\} = \left[\frac{1 + TP^1_{iz}}{1 + TP^0_{iz}} D_L,\ \frac{1 + TP^1_{iz}}{1 + TP^0_{iz}} D_H\right].$$

Protection strictly reduces innovation exactly when the firm innovates at TP^0_{iz} but not at TP^1_{iz}, that is, when D_z lies in the first interval but not the second. Since $(1+TP^1_{iz})/(1+TP^0_{iz}) > 1$ and $D_z \geq D_L$, this requires $D_z < [(1 + TP^1_{iz})/(1 + TP^0_{iz})]\, D_L$. Hence

$$\Delta I^*_{iz} = -1 \quad \Longleftrightarrow \quad D_z \in \left[D_L,\ D^{I,*}_{iz}\right), \qquad D^{I,*}_{iz} = \min\left\{\frac{1 + TP^1_{iz}}{1 + TP^0_{iz}} D_L,\ D_H\right\}.$$

Elsewhere $\Delta I^*_{iz} \geq 0$: for $D_z < D_L$ the firm innovates at neither protection level, so $\Delta I^*_{iz} = 0$; for $D_z \geq D^{I,*}_{iz}$ the firm innovates at both levels, at neither, or at TP^1_{iz} only.

It remains to show that the parity condition removes the region below D_L from the relevant domain. The condition $G_{iz}(1, TP^0_{iz}) \geq F_z > G_{iz}(1, TP^1_{iz})$ states that $\Delta I^*_{iz} = -1$ at $D_z = 1$, so $D_z = 1$ lies in the interval above: $D_L \leq 1 < D^{I,*}_{iz}$. Therefore, on the domain $D_z \geq 1$, the region $[1, D_L)$ is empty, and

$$\Delta I^*_{iz} < 0 \text{ for all } D_z < D^{I,*}_{iz}, \qquad \Delta I^*_{iz} \geq 0 \text{ for all } D_z \geq D^{I,*}_{iz}.$$

■

Appendix D.3.4 Proof of Corollary 2

Proof. By Corollary 1, on the domain $D_z \geq 1$ we have $\Delta I^*_{iz} = -1$ for all $D_z < D^{I,*}_{iz}$ and $\Delta I^*_{iz} \geq 0$ for all $D_z \geq D^{I,*}_{iz}$.

For $D_z < D^{I,*}_{iz}$, the below-threshold argument in the proof of Proposition 2 applies. The condition $\alpha > (1+TP^1_{iz})/(1+TP^0_{iz})$ does not involve D_z, so it holds throughout the interval or nowhere in it; under the hypothesis of the corollary it holds throughout, and $\Delta \log Y_{iz} < 0$ for every $D_z < D^{I,*}_{iz}$.

For $D_z \geq D^{I,*}_{iz}$, the above-threshold argument in the proof of Proposition 2 applies and $\Delta \log Y_{iz} > 0$.

The output effect is therefore strictly negative below $D^{I,*}_{iz}$ and strictly positive at and above it, so $D^{Y,*}_{iz} = D^{I,*}_{iz}$ and the effect of protection on output single-crosses zero from below on $D_z \geq 1$. ■

Appendix D.4 Welfare Appendix

Appendix D.4.1 Numerical Welfare Examples

Consider the welfare change from autarky to free trade:

$$\Delta \log W^{A \to FT} \equiv \log W^{FT} - \log W^{A}.$$

Since $\lambda^A_D = 1$, the welfare change is

$$\Delta \log W^{A \to FT} = \frac{1}{1-\sigma} \log \lambda^{FT}_D + (I^{FT} - I^A) \log \alpha + \log \left(\frac{1 - FI^{FT}}{1 - FI^A} \right).$$

Take

$$\sigma = 4, \qquad \alpha = 2, \qquad F = 0.15.$$

In autarky, the innovation gain is above the fixed cost:

$$G^A_0 = 0.276 > 0.15, \qquad G^A_1 = 0.166 > 0.15.$$

Hence

$$I^A = 1.$$

Close to frontier. Let

$$D = 1.2.$$

Under free trade, the innovation gain is

$$G^{FT} = 0.300 > 0.15,$$

so the firm continues to innovate:

$$I^{FT} = 1.$$

The free-trade domestic expenditure share is

$$\lambda_D^{FT} = 0.664.$$

Therefore

$$\Delta \log W^{A \to FT} = -\frac{1}{3} \log(0.664) = 0.136 > 0.$$

Thus free trade raises welfare for the close-to-frontier product.

Farther from frontier. Now let

$$D = 2.5.$$

Under free trade, the innovation gain is

$$G^{FT} = 0.108 < 0.15,$$

so free trade eliminates innovation:

$$I^{FT} = 0.$$

The free-trade domestic expenditure share is

$$\lambda_D^{FT} = 0.235.$$

The ACR term is

$$-\frac{1}{3} \log(0.235) = 0.483.$$

The innovation/resource term is

$$-\log 2 + \log \left(\frac{1}{1 - 0.15} \right) = -0.531.$$

Therefore

$$\Delta \log W^{A \to FT} = 0.483 - 0.531 = -0.047 < 0.$$

Thus free trade lowers welfare for the farther-from-frontier product.

Appendix D.4.2 Derivation of One-Product Welfare Formula

Welfare is

$$W = \frac{E}{P}.$$

Aggregate income is

$$Y = wL + \Pi,$$

where Π is net profit. Taking logs,

$$d\log Y = \frac{wL}{Y}(d\log w + d\log L) + \frac{\Pi}{Y} d\log \Pi.$$

Without loss of generality, set $w = 1$ as numeraire. In addition, L is fixed, implying $d\log L = 0$:

$$d\log Y = \frac{\Pi}{Y} d\log \Pi.$$

Let λ denote the share of expenditure spent on the domestic variety. CES demand implies

$$\lambda = \left(\frac{p}{P}\right)^{1-\sigma}.$$

Taking logs and rearranging,

$$-\log P = -\log p + \frac{1}{1-\sigma}\log \lambda.$$

Domestic Bertrand pricing implies

$$p = \mu c(I),$$

where

$$\mu = \frac{\varepsilon}{\varepsilon - 1}, \qquad \varepsilon = \sigma - (\sigma - 1)\lambda,$$

and

$$c(I) = \frac{1}{\alpha^I T^{US}}.$$

Therefore,

$$-\log P = -\log \mu + I\log \alpha + \log T^{US} + \frac{1}{1-\sigma}\log \lambda.$$

Operating profits are

$$\Pi^{op} = \frac{\lambda E}{\varepsilon}.$$

Net profits are

$$\Pi = \Pi^{op} - FI.$$

Using $E = Y + B = L + \Pi + B = L + B + \Pi^{op} - FI$, we can write

$$E = L + B - FI + \frac{\lambda E}{\varepsilon},$$

so

$$E = \frac{L + B - FI}{1 - \frac{\lambda}{\varepsilon}}.$$

Combining terms,

$$\log W = \log(L + B - FI) - \log\left(1 - \frac{\lambda}{\varepsilon}\right) - \log\mu + I\log\alpha + \frac{1}{1-\sigma}\log\lambda + \log T^{US}.$$

The markup and profit-income terms cancel because

$$\mu\left(1 - \frac{\lambda}{\varepsilon}\right) = \frac{\varepsilon}{\varepsilon - 1}\left(1 - \frac{\lambda}{\varepsilon}\right) = \frac{\varepsilon - \lambda}{\varepsilon - 1}.$$

Using $\varepsilon = \sigma - (\sigma - 1)\lambda$,

$$\varepsilon - \lambda = \sigma(1 - \lambda), \qquad \varepsilon - 1 = (\sigma - 1)(1 - \lambda).$$

Thus

$$\mu\left(1 - \frac{\lambda}{\varepsilon}\right) = \frac{\sigma}{\sigma - 1},$$

which is constant. Since $d\log T^{US} = 0$,

$$d\log W = \frac{1}{1-\sigma}d\log\lambda + dI\log\alpha + d\log(L + B - FI).$$

Finally,

$$d\log(L + B - FI) = -\frac{F}{L + B - FI}dI.$$

Therefore,

$$d\log W = \frac{1}{1-\sigma}d\log\lambda + dI\left(\log\alpha - \frac{F}{L + B - FI}\right).$$

Appendix D.4.3 Derivation of Multiproduct Welfare Formula

With Cobb-Douglas preferences across products,

$$P = \prod_z P_z^{\chi_z},$$

so

$$d\log P = \sum_z \chi_z d\log P_z.$$

Within each product,

$$\lambda_z = \left(\frac{p_z}{P_z}\right)^{1-\sigma}.$$

Therefore,

$$d\log P_z = d\log p_z - \frac{1}{1-\sigma} d\log \lambda_z.$$

Domestic Bertrand pricing gives

$$p_z = \mu_z c_z(I_z), \qquad c_z(I_z) = \frac{1}{\alpha^{I_z} T_z^{US}}.$$

Since $d\log T_z^{US} = 0$,

$$d\log p_z = d\log \mu_z - dI_z \log \alpha.$$

Thus

$$d\log P_z = d\log \mu_z - dI_z \log \alpha - \frac{1}{1-\sigma} d\log \lambda_z.$$

Welfare is $W = E/P$, so

$$d\log W = d\log E - d\log P.$$

Aggregate income before fixed costs is

$$Y = wL + \sum_z \Pi_z^{op}.$$

With fixed costs paid in units of the final good,

$$E = Y + B - \sum_z F I_z.$$

With $w = 1$ as numeraire, fixed L, and exogenous B,

$$d\log E = \sum_z \frac{\Pi_z^{op}}{E} d\log \Pi_z^{op} - \sum_z \frac{F}{E} dI_z.$$

Using

$$\frac{\Pi_z^{op}}{E} = \frac{\Pi_z^{op}}{E_z}\frac{E_z}{E} = \chi_z \frac{\Pi_z^{op}}{E_z},$$

and substituting the price-index expression gives

$$
\begin{aligned}
d\log W = & \underbrace{\sum_z \chi_z \frac{1}{1-\sigma} d\log\lambda_z}_{\text{ACR term}} \\
& + \underbrace{\sum_z dI_z \left[\chi_z \log\alpha - \frac{F}{E}\right]}_{\text{innovation effect}} \\
& - \underbrace{\sum_z \chi_z d\log\mu_z}_{\text{procompetitive effect}} + \underbrace{\sum_z \chi_z \frac{\Pi_z^{op}}{E_z} d\log\Pi_z^{op}}_{\text{rent effect}}.
\end{aligned}
$$

Appendix D.4.4 Tariff-Revenue Extension

Single Product Case:

When the shock is a tariff rather than an iceberg trade-cost shock, tariff revenue enters expenditure. Let t denote the tariff and $1-\lambda$ the import share. Tariff revenue is

$$TR = \frac{t}{1+t}(1-\lambda)E.$$

Thus

$$\frac{TR}{E} = \frac{t}{1+t}(1-\lambda).$$

Accounting for tariff revenues therefore adds the term

$$-d\log\left[1 - \frac{t}{1+t}(1-\lambda)\right].$$

The one-product welfare formula becomes

$$d\log W = \frac{1}{1-\sigma} d\log\lambda + dI\left(\log\alpha - \frac{F}{L+B-FI}\right) - d\log\left[1 - \frac{t}{1+t}(1-\lambda)\right].$$

The last term is the tariff-revenue effect.

Multi Product Case:

In the multiproduct case, tariff revenue is

$$TR = E\sum_z \chi_z \frac{t_z}{1+t_z}(1-\lambda_z).$$

Thus

$$\frac{TR}{E} = \sum_z \chi_z \frac{t_z}{1+t_z}(1-\lambda_z).$$

Therefore, accounting for tariff revenue adds the following term to the multiproduct welfare formula:

$$-d\log\left[1-\sum_z \chi_z \frac{t_z}{1+t_z}(1-\lambda_z)\right].$$

Thus,

$$\begin{aligned} d\log W = & \underbrace{\sum_z \chi_z \frac{1}{1-\sigma} d\log\lambda_z}_{\text{ACR term}} \\ & \underbrace{+\sum_z dI_z\left[\chi_z \log\alpha - \frac{F}{E}\right]}_{\text{innovation effect}} \\ & - \underbrace{\sum_z \chi_z d\log\mu_z}_{\text{procompetitive effect}} + \underbrace{\sum_z \chi_z \frac{\Pi_z^{op}}{E_z} d\log\Pi_z^{op}}_{\text{rent effect}} \\ & \underbrace{- d\log\left[1-\sum_z \chi_z \frac{t_z}{1+t_z}(1-\lambda_z)\right]}_{\text{tariff-revenue effect}}. \end{aligned}$$

Appendix D.5 Continuous Model For Comparative Statics

This appendix presents a continuous-innovation version of the model in 3. Here the US firm chooses a continuous innovation effort rather than making a discrete upgrade decision.

Appendix D.5.1 Setup

A US firm and a UK firm compete in the US market for product z à la Bertrand. US expenditure on product z is E_z, which is fixed in this partial-equilibrium model. Demand is CES with elasticity $\sigma > 1$. If the US and UK prices are p_z^{US} and p_z^{UK}, the US expenditure share is

$$\lambda_z = \frac{(p_z^{US})^{1-\sigma}}{(p_z^{US})^{1-\sigma} + (p_z^{UK})^{1-\sigma}}.$$

Let US and UK technologies be T_z^{US} and T_z^{UK}. Define distance to the frontier as

$$D_z \equiv \frac{T_z^{UK}}{T_z^{US}}.$$

The tariff raises the delivered marginal cost of the UK firm in the US market:

$$c_z^{UK} = \frac{1+t_z}{T_z^{UK}}.$$

The US firm chooses innovation effort $n_z \geq 0$. Innovation raises US technology from T_z^{US} to

$$T_z^{US}\exp(n_z^{\alpha}), \qquad 1 < \alpha < 2.$$

Thus, the US unit cost is

$$c_z^{US}(n_z) = \frac{1}{T_z^{US}\exp(n_z^{\alpha})}.$$

Define relative US competitiveness as

$$r_z(n_z, t_z) \equiv \frac{c_z^{UK}}{c_z^{US}(n_z)} = \frac{(1+t_z)\exp(n_z^{\alpha})}{D_z}.$$

Appendix D.5.2 Pricing

Under Bertrand competition with CES demand, the perceived own-price elasticity of the US firm is

$$\varepsilon_z^{US} = \sigma - (\sigma-1)\lambda_z.$$

The perceived own-price elasticity of the UK firm is

$$\varepsilon_z^{UK} = 1 + (\sigma-1)\lambda_z.$$

Therefore, optimal prices satisfy

$$p_z^{US} = \frac{\sigma-(\sigma-1)\lambda_z}{(\sigma-1)(1-\lambda_z)}c_z^{US},$$

and

$$p_z^{UK} = \frac{1+(\sigma-1)\lambda_z}{(\sigma-1)\lambda_z}c_z^{UK}.$$

Combining the CES share equation with the Bertrand pricing equations, the US expenditure share $\lambda_z(r_z)$ is implicitly defined by

$$\frac{\lambda_z(r_z)}{1-\lambda_z(r_z)} = \left[\frac{[1+(\sigma-1)\lambda_z(r_z)][1-\lambda_z(r_z)]}{\lambda_z(r_z)[\sigma-(\sigma-1)\lambda_z(r_z)]}r_z\right]^{\sigma-1}.$$

Operating profit is

$$\Pi_z(r_z) = E_z\frac{\lambda_z(r_z)}{\sigma-(\sigma-1)\lambda_z(r_z)}.$$

Appendix D.5.3 Innovation Problem

Innovation effort is costly. Let

$$C_z(n_z) = \kappa_{0z}n_z + \frac{\kappa_{1z}}{2}n_z^2, \qquad \kappa_{0z} > 0, \qquad \kappa_{1z} > 0.$$

The firm chooses innovation effort to solve

$$n_z^* \in \arg\max_{n_z \geq 0}\left\{\Pi_z\left(\frac{(1+t_z)\exp(n_z^{\alpha})}{D_z}\right) - \kappa_{0z}n_z - \frac{\kappa_{1z}}{2}n_z^2\right\}.$$

For an interior optimum, the first-order condition is

$$\frac{\partial \Pi_z(r_z^*)}{\partial \log r_z}\alpha(n_z^*)^{\alpha-1} = \kappa_{0z} + \kappa_{1z} n_z^*,$$

where

$$r_z^* = \frac{(1+t_z)\exp((n_z^*)^\alpha)}{D_z}.$$

Let

$$\Omega_z \equiv \kappa_{1z} - \frac{\partial \Pi_z(r_z^*)}{\partial \log r_z}\alpha(\alpha-1)(n_z^*)^{\alpha-2} - \frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}\alpha^2(n_z^*)^{2\alpha-2}.$$

Throughout this appendix, I focus on regular interior optima, meaning optima with $n_z^* > 0$ and $\Omega_z > 0$.[39]

Appendix D.5.4 Comparative Statics

Differentiating the first-order condition with respect to $\log(1+t_z)$ gives

$$\frac{dn_z^*}{d\log(1+t_z)} = \frac{\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}\alpha(n_z^*)^{\alpha-1}}{\Omega_z}.$$

Differentiating the first-order condition with respect to $\log D_z$ gives

$$\frac{dn_z^*}{d\log D_z} = -\frac{\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}\alpha(n_z^*)^{\alpha-1}}{\Omega_z}.$$

US sales are

$$Y_z = E_z\lambda_z\left(\frac{(1+t_z)\exp((n_z^*)^\alpha)}{D_z}\right).$$

Therefore,

$$\frac{d\log Y_z}{d\log(1+t_z)} = \frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z}\left[1 + \alpha(n_z^*)^{\alpha-1}\frac{dn_z^*}{d\log(1+t_z)}\right].$$

Substituting the innovation response and using the first-order condition gives

$$\frac{d\log Y_z}{d\log(1+t_z)} = \frac{\frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z}}{\Omega_z}\left[\kappa_{1z}(2-\alpha) - \frac{\kappa_{0z}(\alpha-1)}{n_z^*}\right].$$

Define

$$\bar{n}_z \equiv \frac{\kappa_{0z}(\alpha-1)}{\kappa_{1z}(2-\alpha)}.$$

[39] A sufficient condition for the optimum to be interior is that there exists some $\tilde{n}_z > 0$ such that

$$\Pi_z\left(\frac{(1+t_z)\exp(\tilde{n}_z^\alpha)}{D_z}\right) - \kappa_{0z}\tilde{n}_z - \frac{\kappa_{1z}}{2}\tilde{n}_z^2 > \Pi_z\left(\frac{1+t_z}{D_z}\right).$$

Appendix D.5.5 Main Propositions

I now state the main comparative statics results. The first proposition characterizes the innovation response. The second characterizes the output response. The third gives the local monotonicity result in distance to the frontier.

Proposition A1: There exists a threshold D_z^I such that for all $D_z \geq D_z^I$, protection weakly increases innovation $\left(\frac{dn_z^*}{d\log(1+t_z)} \geq 0\right)$. In addition, for some parameter values, there can exist products with $D_z < D_z^I$ for which protection reduces innovation $\left(\frac{dn_z^*}{d\log(1+t_z)} < 0\right)$.

Suppose there exists a unique value $\bar{r}_z$ such that

$$\frac{\partial^2 \Pi_z(\bar{r}_z)}{\partial(\log r_z)^2} = 0,$$

with

$$\frac{\partial^2 \Pi_z(r_z)}{\partial(\log r_z)^2} > 0 \quad \text{for} \quad r_z < \bar{r}_z,$$

and

$$\frac{\partial^2 \Pi_z(r_z)}{\partial(\log r_z)^2} < 0 \quad \text{for} \quad r_z > \bar{r}_z.$$

Consider a regular interval of distance-to-frontier values over which $r_z^*(D_z)$ is strictly decreasing in D_z and crosses $\bar{r}_z$ once. Then there exists a unique threshold D_z^I such that protection weakly increases innovation for $D_z \geq D_z^I$ and reduces innovation for $D_z < D_z^I$.

Proof. From the comparative statics above,

$$\frac{dn_z^*}{d\log(1+t_z)} = \frac{\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}\alpha(n_z^*)^{\alpha-1}}{\Omega_z}.$$

Since the optimum is regular,

$$\Omega_z > 0.$$

Since $\alpha > 1$ and $n_z^* > 0$,

$$\alpha(n_z^*)^{\alpha-1} > 0.$$

Therefore the sign of the innovation response is the sign of

$$\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}.$$

By assumption, this term is positive when $r_z^* < \bar{r}_z$ and negative when $r_z^* > \bar{r}_z$. Since $r_z^*(D_z)$ is strictly decreasing in D_z and crosses $\bar{r}_z$ once, there exists a unique D_z^I such that

$$r_z^*(D_z^I) = \bar{r}_z.$$

For $D_z > D_z^I$, $r_z^*(D_z) < \bar{r}_z$, so

$$\frac{dn_z^*}{d\log(1+t_z)} > 0.$$

For $D_z < D_z^I$, $r_z^*(D_z) > \bar{r}_z$, so

$$\frac{dn_z^*}{d\log(1+t_z)} < 0.$$

At $D_z = D_z^I$, the innovation response is zero. Thus protection weakly increases innovation above the threshold and reduces innovation below it. ■

Proposition A2: There exists a threshold D_z^Y such that for all $D_z \geq D_z^Y$, protection raises output $\left(\frac{d\log Y_z}{d\log(1+t_z)} > 0\right)$. In addition, for some parameter values, there can exist products with $D_z < D_z^Y$ for which protection reduces output $\left(\frac{d\log Y_z}{d\log(1+t_z)} < 0\right)$.

Consider a regular interval $[D_L, D_H]$ over which

$$\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2} < 0.$$

Suppose

$$n_z^*(D_L) < \bar{n}_z < n_z^*(D_H).$$

Then there exists a unique threshold $D_z^Y \in (D_L, D_H)$ such that protection reduces output for $D_z < D_z^Y$ and raises output for $D_z > D_z^Y$.

Proof. Over this interval,

$$\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2} < 0.$$

Therefore,

$$\frac{dn_z^*}{d\log D_z} = -\frac{\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}\alpha(n_z^*)^{\alpha-1}}{\Omega_z} > 0.$$

Thus $n_z^*(D_z)$ is strictly increasing in distance to frontier over $[D_L, D_H]$.

The output elasticity is

$$\frac{d\log Y_z}{d\log(1+t_z)} = \frac{\frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z}}{\Omega_z}\left[\kappa_{1z}(2-\alpha) - \frac{\kappa_{0z}(\alpha-1)}{n_z^*}\right].$$

Because

$$\frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z} > 0 \qquad \text{and} \qquad \Omega_z > 0,$$

the sign of the output response is the sign of

$$\kappa_{1z}(2-\alpha) - \frac{\kappa_{0z}(\alpha-1)}{n_z^*}.$$

This expression is negative when

$$n_z^* < \bar{n}_z$$

and positive when

$$n_z^* > \bar{n}_z.$$

Since $n_z^*(D_z)$ is strictly increasing and

$$n_z^*(D_L) < \bar{n}_z < n_z^*(D_H),$$

there exists a unique $D_z^Y \in (D_L, D_H)$ such that

$$n_z^*(D_z^Y) = \bar{n}_z.$$

Hence protection reduces output below D_z^Y and raises output above D_z^Y. ■

Proposition A3: There exists a range such that the effect of protection on output is monotonic in distance to frontier.

Under the conditions of Proposition A2, at the threshold $D_z = D_z^Y$,

$$\frac{\partial}{\partial \log D_z}\left[\frac{d\log Y_z}{d\log(1+t_z)}\right] > 0.$$

Therefore, in a neighborhood of D_z^Y, the effect of protection on output is increasing in distance to the frontier.

Proof. At $D_z = D_z^Y$,

$$n_z^*(D_z^Y) = \bar{n}_z.$$

Therefore,

$$\kappa_{1z}(2-\alpha) - \frac{\kappa_{0z}(\alpha-1)}{n_z^*} = 0.$$

Differentiating the output elasticity with respect to $\log D_z$ and evaluating at $D_z = D_z^Y$ gives

$$\frac{\partial}{\partial \log D_z}\left[\frac{d\log Y_z}{d\log(1+t_z)}\right] = \frac{\frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z}}{\Omega_z}\frac{\kappa_{0z}(\alpha-1)}{(n_z^*)^2}\frac{dn_z^*}{d\log D_z}.$$

Each term on the right-hand side is positive. Therefore,

$$\frac{\partial}{\partial \log D_z}\left[\frac{d\log Y_z}{d\log(1+t_z)}\right] > 0.$$

By continuity, the same inequality holds in a neighborhood of D_z^Y. ■

Appendix D.6 Heterogeneous Firms Model

Here I present a heterogeneous-firms version of the baseline model in the text. This is close in spirit to Bustos (2011), who augments a Melitz model with a technology-adoption decision. The key difference is that demand is Kimball rather than CES, which allows for variable markups and residual demand elasticities that vary with firm size.

I show that this model shares the key prediction in Bustos (2011): after an export-market-access shock, firms in the upper-middle part of the productivity distribution may upgrade technology. However, the model has an additional implication. Because the incremental return to innovation need not be monotone in productivity under Kimball demand, firms at the top of the productivity distribution may reduce innovation after export market access improves. Therefore, the aggregate effect on innovation depends on the relative mass of firms induced to upgrade versus firms induced to stop upgrading.

The focus of this appendix is on heterogeneity across firms **within** a product of a given distance to frontier. However, under additional restrictions, this heterogeneous firms model can also generate the same aggregate predictions in the main text: protection reduces output and innovation for close to frontier items, yet increases output and innovation for sufficiently behind the frontier.

Environment. Consider product z. There is a continuum of potential domestic firms indexed by ω. A firm draws productivity

$$\varphi_\omega > 0$$

from distribution $H_z(\varphi)$, with density $h_z(\varphi)$. The support is

$$\varphi_\omega \in [\underline{\varphi}_z, \infty).$$

A potential entrant pays entry cost f_{ez}, draws φ_ω, and then chooses whether to operate and whether to adopt an improved technology.

Let

$$I_{z\omega} \in \{0, 1\}$$

denote the firm's technology choice. If $I_{z\omega} = 0$, the firm's technology is $T_z^{US}\varphi_\omega$. If $I_{z\omega} = 1$, its technology is

$$\alpha T_z^{US}\varphi_\omega, \qquad \alpha > 1.$$

Adoption requires fixed cost $F_z > 0$.

Distance to the foreign frontier is

$$D_z \equiv \frac{T_z^{UK}}{T_z^{US}}.$$

Thus, holding T_z^{UK} fixed, larger D_z means the domestic productivity distribution is shifted farther left relative to the foreign frontier.

Let m index markets. The domestic market is $m = d$, and the export market is $m = x$. Serving market m requires fixed cost f_{zm}. Conditional on serving market m, the firm's marginal cost is

$$c_{zm\omega}(I_{z\omega}) = \frac{w_z \tau_{zm}}{\alpha^{I_{z\omega}} T_z^{US} \varphi_\omega},$$

where w_z is the wage and τ_{zm} is the iceberg trade cost.

Kimball demand and pricing. In market m, let

$$x_{zm\omega} \equiv \frac{q_{zm\omega}}{Q_{zm}}.$$

Preferences are Kimball:

$$\int_{\Omega_{zm}} \Upsilon(x_{zm\omega})d\omega = 1.$$

Inverse demand is

$$p_{zm\omega} = B_{zm}\Upsilon'(x_{zm\omega}),$$

where each atomistic firm takes B_{zm} and Q_{zm} as given.

Use the Klenow-Willis parametrization:

$$\Upsilon'(x) = \frac{\bar{\sigma}-1}{\bar{\sigma}} \exp\left(\frac{1-x^{\theta/\bar{\sigma}}}{\theta}\right), \qquad \bar{\sigma} > 1, \quad \theta > 0.$$

Define

$$\widetilde{B}_{zm} \equiv B_{zm}\frac{\bar{\sigma}-1}{\bar{\sigma}}, \qquad u_{zm\omega} \equiv x_{zm\omega}^{\theta/\bar{\sigma}}.$$

Then

$$p_{zm\omega} = \widetilde{B}_{zm} \exp\left(\frac{1-u_{zm\omega}}{\theta}\right),$$

and

$$q_{zm\omega} = Q_{zm}u_{zm\omega}^{\bar{\sigma}/\theta}.$$

The local demand elasticity is

$$\varepsilon_{zm\omega} = \frac{\bar{\sigma}}{u_{zm\omega}}.$$

The firm's pricing first-order condition is

$$p_{zm\omega} = \frac{\varepsilon_{zm\omega}}{\varepsilon_{zm\omega}-1}c_{zm\omega}.$$

Using $\varepsilon_{zm\omega} = \bar{\sigma}/u_{zm\omega}$, this is equivalent to

$$\frac{c_{zm\omega}}{\widetilde{B}_{zm}} = \exp\left(\frac{1-u_{zm\omega}}{\theta}\right)\frac{\bar{\sigma}-u_{zm\omega}}{\bar{\sigma}}.$$

Define

$$m(u) \equiv \exp\left(\frac{1-u}{\theta}\right)\frac{\bar{\sigma}-u}{\bar{\sigma}}.$$

Then

$$\frac{c_{zm\omega}}{\widetilde{B}_{zm}} = m(u_{zm\omega}).$$

Since

$$\frac{d\log m(u)}{du} = -\frac{1}{\theta} - \frac{1}{\bar{\sigma}-u} < 0,$$

each marginal cost pins down a unique $u_{zm\omega}$. Higher productivity raises $u_{zm\omega}$.

Firm revenue is

$$r_{zm\omega} = p_{zm\omega}q_{zm\omega}.$$

Variable profit is

$$\pi^v_{zm\omega} = \frac{r_{zm\omega}}{\varepsilon_{zm\omega}}.$$

Substituting the KW demand objects,

$$\pi^v_{zm\omega} = \frac{\widetilde{B}_{zm} Q_{zm}}{\bar{\sigma}} \exp\left(\frac{1 - u_{zm\omega}}{\theta}\right) u_{zm\omega}^{\bar{\sigma}/\theta+1}.$$

Define log competitiveness in market m by

$$\ell_{zm\omega} \equiv \log\left(\frac{\widetilde{B}_{zm} T_z^{US} \varphi_\omega}{w_z \tau_{zm}}\right).$$

Using $D_z = T_z^{UK}/T_z^{US}$, this can also be written as

$$\ell_{zm\omega} = \log\varphi_\omega - \log D_z + \log\left(\frac{\widetilde{B}_{zm} T_z^{UK}}{w_z \tau_{zm}}\right).$$

Thus, a larger D_z shifts domestic firms toward lower competitiveness.

Innovation raises log competitiveness from $\ell_{zm\omega}$ to

$$\ell_{zm\omega} + \log\alpha.$$

Let

$$g_{zm}(\ell)$$

denote maximized variable profit in market m as a function of log competitiveness. Then

$$\pi^v_{zm\omega}(0) = g_{zm}(\ell_{zm\omega}),$$

and

$$\pi^v_{zm\omega}(1) = g_{zm}(\ell_{zm\omega} + \log\alpha).$$

Technology adoption. Let s_z denote the aggregate share of domestic firms in product z that adopt the improved technology. Let

$$\Delta_z(\varphi; s_z) \equiv \pi^v_z(\alpha\varphi; s_z) - \pi^v_z(\varphi; s_z)$$

denote the gross gain from innovation for a firm with productivity φ, taking the aggregate adoption state s_z as given. A firm adopts iff

$$\Delta_z(\varphi; s_z) \geq F_z.$$

I assume aggregate adoption does not increase the private gain from adoption:

$$\frac{\partial\Delta_z(\varphi; s_z)}{\partial s_z} \leq 0 \qquad \forall\varphi,\ s_z \in [0,1]. \tag{A1}$$

This condition says that adoption choices are strategic substitutes: when more firms adopt, the private incremental gain from adopting weakly falls. This assumption guarantees a unique equilibrium as it rules out multiple self-fulfilling adoption equilibria generated by aggregate innovation feedback.

Given s_z, define the set of adopters

$$\mathcal{I}_z(s_z) = \{\varphi : \Delta_z(\varphi; s_z) \geq F_z\}.$$

The equilibrium adoption share satisfies

$$s_z = \int_{\mathcal{I}_z(s_z)} dH_z(\varphi).$$

Under (A1), the right-hand side is weakly decreasing in s_z. The left-hand side is strictly increasing in s_z. Therefore the fixed point is unique.

Proposition: Lower Innovation Cutoff. Suppose $\Delta_z(\varphi)$ is continuous,

$$\Delta_z(\underline{\varphi}_z) < F_z,$$

and there exists some $\tilde{\varphi}_z > \underline{\varphi}_z$ such that

$$\Delta_z(\tilde{\varphi}_z) > F_z.$$

Then there exists a lower innovation cutoff φ_{Lz} such that firms below φ_{Lz} do not innovate, while some firms above φ_{Lz} innovate.

Proof. Define

$$\varphi_{Lz} \equiv \inf\{\varphi \geq \underline{\varphi}_z : \Delta_z(\varphi) \geq F_z\}.$$

The set is nonempty because $\Delta_z(\tilde{\varphi}_z) > F_z$. Since $\Delta_z(\underline{\varphi}_z) < F_z$, we have

$$\varphi_{Lz} > \underline{\varphi}_z.$$

By continuity,

$$\Delta_z(\varphi_{Lz}) = F_z.$$

By construction, for all $\varphi < \varphi_{Lz}$,

$$\Delta_z(\varphi) < F_z,$$

so firms below φ_{Lz} do not innovate. Since $\Delta_z(\tilde{\varphi}_z) > F_z$, some firms above φ_{Lz} innovate. □

Proposition: Upper Innovation Cutoff under Kimball. Under KW-Kimball demand, the innovation gain is not necessarily increasing in productivity. In particular, for sufficiently high-productivity firms,

$$\frac{d\Delta_z(\varphi)}{d\varphi} < 0.$$

If, in addition,

$$\Delta_z(\varphi) < F_z \quad \text{for sufficiently high } \varphi,$$

and some intermediate-productivity firms innovate, then there exists an upper innovation cutoff φ_{Hz} such that firms above φ_{Hz} do not innovate. Under the additional restriction that there are no further crossings, firms innovate only on an intermediate interval:

$$\varphi_\omega \in [\varphi_{Lz}, \varphi_{Hz}].$$

Proof. Let $s = \log \varphi$ and $a = \log \alpha > 0$. Write variable profits as $g_z(s)$. Then

$$\Delta_z(s) = g_z(s + a) - g_z(s).$$

Differentiating,

$$\Delta_z'(s) = g_z'(s + a) - g_z'(s) = \int_s^{s+a} g_z''(r)dr.$$

Under KW-Kimball, profits are convex at low competitiveness and concave at high competitiveness. Therefore, for sufficiently high s,

$$g_z''(r) < 0 \quad \text{for all } r \in [s, s + a],$$

so

$$\Delta_z'(s) < 0.$$

Thus the gain from innovation falls for sufficiently high-productivity firms.

Now suppose some intermediate firms innovate but sufficiently high-productivity firms do not. Define

$$\varphi_{Hz} \equiv \sup\{\varphi : \Delta_z(\varphi) \geq F_z\}.$$

Continuity implies

$$\Delta_z(\varphi_{Hz}) = F_z.$$

By construction, firms above φ_{Hz} do not innovate. If there are no additional crossings between φ_{Lz} and φ_{Hz}, then firms innovate iff

$$\varphi_\omega \in [\varphi_{Lz}, \varphi_{Hz}].$$

□

Market-access comparative static. Let a_z denote market access. For export access, a_z rises when foreign tariffs faced by domestic exporters fall. For import protection, a_z rises when tariffs faced by foreign competitors in the domestic market rise.

At the lower innovation margin, market access raises the gain from innovation:

$$\Delta_{z,a}(\varphi_{Lz}) > 0.$$

If $\Delta_{z,\varphi}(\varphi_{Lz}) > 0$, then

$$\frac{d\varphi_{Lz}}{da_z} = -\frac{\Delta_{z,a}(\varphi_{Lz})}{\Delta_{z,\varphi}(\varphi_{Lz})} < 0.$$

Thus improved market access induces some middle-productivity firms to innovate.

At the upper innovation margin, market access can reduce the gain from innovation:

$$\Delta_{z,a}(\varphi_{Hz}) < 0.$$

If $\Delta_{z,\varphi}(\varphi_{Hz}) < 0$, then

$$\frac{d\varphi_{Hz}}{da_z} = -\frac{\Delta_{z,a}(\varphi_{Hz})}{\Delta_{z,\varphi}(\varphi_{Hz})} < 0.$$

Thus improved market access can induce some high-productivity firms to stop innovating.

If the adoption set is $[\varphi_{Lz}, \varphi_{Hz}]$, aggregate innovation is

$$I_z = H_z(\varphi_{Hz}) - H_z(\varphi_{Lz}).$$

Therefore,

$$\frac{dI_z}{da_z} = h_z(\varphi_{Hz})\frac{d\varphi_{Hz}}{da_z} - h_z(\varphi_{Lz})\frac{d\varphi_{Lz}}{da_z}.$$

The lower-margin term is positive, while the upper-margin term is negative. Hence the aggregate innovation response is ambiguous.

Relation to Bustos. The lower-cutoff force is the Bustos (2011) force: improved export-market access induces firms near the technology-adoption margin to upgrade. The Kimball model adds the possibility of an upper innovation cutoff. Therefore, market access can also induce sufficiently productive firms to stop innovating, making the aggregate innovation response depend on the relative mass of firms at the two margins.